FACULTY OF COMPUTER SCIENCE & INFORMATION COMPUTING TECHNOLOGY

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PROJECT REPORT

ONLINE TWO PLAYER POKER GAME
MIND POKER

CALVIN NG ZHI QI

2290228-DCS

Report submitted in partial fulfilment of the requirements for

DIPLOMA IN COMPUTER SCIENCE

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DECLARATION

I hereby declare that this project report is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at New Era University College or other institutions.

Signature : ____________________

Name : CALVIN NG ZHI QI

Student ID : 2290228-DCS

Date : 24^{th} SEPTEMBER 2024

24^{th} SEPTEMBER 2024
CALVIN NG ZHI QI
2290228-DCS

CERTIFICATION OF APPROVAL

ONLINE TWO PLAYER POKER CARD GAME
MIND POKER

BY

CALVIN NG ZHI QI

This project was prepared under the direction of project coordinator, Ms Suhailah and it has been approved by project supervisor Ms Suhailah. It was submitted to the Faculty of Information & Computing Technology and was accepted in partial fulfilment of the requirement for the Diploma of Computer Science.

Approved by

Miss Suhailah
Project Supervisor

Miss Suhailah
Project Coordinator

24th SEPTEMBER 2024

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LIST OF ABBREVIATIONS

The following abbreviations have been used to label grammatical morphemes in the glosses in this dissertation.

DFD	Data Flow Diagram
FAB	Floating Action Button
JSON	Javascript Object Notation
POS	Point of Sales
TNB	Tenaga Nasional Berhad
SDLC	Software Development Life Cycle
SQFT	Square Footage
SYABAS	Syarikat Bekalan Air Selangor
URL	Uniform Resource Locator

ABSTRACT

This project is about developing an online game called Mind Poker. This project is a game that emphasizes intelligence, mathematics, memory and mental strength. This game allows two players to play against each other online. In addition, the player's strategy will affect the development of the game in the later stages, and the difficulty will increase over time.

CHAPTER 1

1 INTRODUCTION

1.1 Background of the Research

In today's fast-paced and intellectually demanding world, there is a growing interest in games that challenge not just reflexes but also cognitive abilities. This project aims to develop an innovative online game called Mind Poker. Mind Poker is designed to test and enhance players' intelligence, mathematical skills, memory, and mental strength. Unlike traditional games that rely heavily on luck or simple tactics, Mind Poker requires players to engage in strategic thinking and adapt their strategies as the game progresses.

Mind Poker offers a unique multiplayer experience where players can compete against each other online, pushing their cognitive limits in a competitive yet intellectually stimulating environment. As players advance through the game, the difficulty increases, demanding more sophisticated strategies and deeper concentration. This gradual increase in complexity ensures that players are constantly challenged and engaged, making Mind Poker an ideal game for those who enjoy mental exercises and strategic planning.

The development of Mind Poker integrates advanced online technologies to create a seamless and interactive gaming experience. Players' decisions and strategies will significantly influence the game's outcome, promoting a deeper level of engagement and playability. Whether you are a fan of puzzles, strategy games, or simply looking for a mentally stimulating pastime, Mind Poker is designed to provide a rewarding and challenging experience that sharpens your mind and keeps you entertained.

1.2 Problem Statement

In the current landscape of digital entertainment, most games focus heavily on luck or basic tactical skills, often neglecting the cognitive challenges that appeal to more intellectually inclined players. Traditional poker, while strategic, largely depends on chance, making it less appealing to those who enjoy more deterministic and skill-based gameplay.

According to studies on cognitive development, games that stimulate mathematical thinking, memory, and strategic planning can significantly enhance mental acuity. However, there is a lack of such games in the online multiplayer sphere. Mind Poker addresses this gap by providing a online platform where players can engage in a game that requires deep thinking, strategic adjustments, and mental endurance.

The need for a game like Mind Poker is further highlighted by the increasing interest in brain-training games and applications. These games not only entertain but also contribute to mental health by keeping the mind active and sharp. Mind Poker combines the excitement of competitive gaming with the benefits of cognitive exercises, offering a unique and engaging experience for players seeking more than just entertainment.

1.3 Project Objectives

Below are the objectives for these studies:

1. To develop an online multiplayer game that emphasizes intelligence, mathematics, memory, and mental strength where player strategies significantly influence the outcome, enhancing engagement and playability. To create a game environment where player strategies significantly influence the outcome, enhancing engagement and playability.

2. To implement advanced online technologies for a seamless and interactive gaming experience, with progressively increasing difficulty to continuously challenge and engage players.

1.4 Project Scope

The primary target audience for Mind Poker encompasses a diverse range of players, including strategy game enthusiasts, individuals interested in cognitive development, and competitive gamers seeking skill-based challenges. The game is designed to appeal to those who enjoy mental exercises and strategic planning. Mind Poker requires a download for optimal performance and to fully immerse players in the game.

1.5 Project Justification

Below is the significance of the project to be conducted.

1.5.1 Cognitive Development

Mind Poker provides a platform that challenges players' mental faculties, promoting cognitive growth through strategic gameplay and mathematical problem-solving.

1.5.2 Unique Gameplay Experience

The game offers a unique blend of poker mechanics with intellectual challenges, setting it apart from traditional card games that rely heavily on chance.

1.5.3 High playability

The dynamic nature of player strategies and increasing difficulty levels ensure high playability, keeping players engaged and motivated to improve their skills.

1.5.4 Competitive Edge

Mind Poker encourages a competitive spirit among players, fostering a community of like-minded individuals who value intellectual prowess and strategic thinking.

1.6 Project Constraint

There are some of the constraints and limitations with this project.

1.6.1 Complex project

Several databases are needed in this project. During the design phase, the connection between each database must be check to ensure it is connected correctly to each other so that there are no problems during the implementation phase.

1.6.2 Time and skills

The most challenging part of this project is the implementation process, where I need to have good knowledge of programming. Therefore, I need to invest my time in learning the programming language used in this project to enhance my skills and reduce the time I used to solve errors during the implementation and testing phase.

1.7 Project Timeline

The table below shows the Gantt chart and milestone for this research.

Table 1.1 Gantt Chart for the research

Week/task	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Find Literature Review														
Find Problem Statement														
Research Title														
Research Methodology														
Planning														
Design														
Develop														
Report														
Presentations														

The table is a Gantt chart that outlines the timeline for completing various tasks related to a research project over a 14-week period.

The research project begins with a literature review, scheduled from weeks 1 to 3. Simultaneously, the problem statement is developed from weeks 2 to 4. The research title is finalized in week 4, followed by the development of the research methodology from weeks 5 to 7. Planning is conducted from weeks 6 to 8, overlapping with the design phase that spans weeks 7 to 9. The development of the research project takes place from weeks 8 to 10. The report writing phase is planned for weeks 10 to 12, and the project concludes with presentations in week 14. This timeline ensures a structured approach, with specific tasks designated to particular weeks, some of which overlap to maximize efficiency and ensure timely project completion.

1.8 Report Organization

The report is organized into seven chapters, each addressing critical aspects of the project development process. Chapter 1 introduces the project, outlining its background, problem statement, objectives, scope, justifications, and structure of the report. Chapter 2 provides a literature review, discussing related systems, techniques, and technologies, along with a critical analysis of current problems and their justifications. Chapter 3 details the project methodology, including the development lifecycle and the rationale behind the chosen methodology. Chapter 4 focuses on project planning and analysis, covering requirements gathering, hardware and software needs, and both functional and non-functional requirements. Chapter 5 presents the project design, incorporating various design elements such as flowcharts, architecture, and diagrams. Chapter 6 describes the project development process, user manual, and types of testing conducted. Finally, Chapter 7 concludes the report with a project summarization and suggestions for future enhancements.

CHAPTER 2

2 LITERATURE REVIEW

2.1 Introduction

A literature review is a review of the existing research of the chosen topic. The goal of the literature review is to develop a conceptual framework of the subject area and to evaluate models and case studies to support the proposed system. As for this thesis, several research articles and existing systems have been used as references to help in developing this system.

2.2 Overview of Poker Game

Poker is one of the most popular card games worldwide, known for its combination of skill, strategy, and chance. It involves players betting on the strength of their hands, with the goal of either holding the best hand or convincing others to fold. There are numerous variations of poker, such as Texas Hold'em, Omaha, and Seven-Card Stud, each with its own set of rules and strategies.

According to Jones (2017), poker's popularity can be attributed to its unique blend of luck and skill, where players must make strategic decisions based on incomplete information. This dynamic creates a competitive environment where psychological tactics, such as bluffing, are as important as the cards themselves.

Furthermore, as highlighted by Smith (2019), poker is not just a game of chance but also requires a deep understanding of probability, game theory, and human behavior. The ability to read opponents and anticipate their moves can often determine the outcome of a game, making poker a mentally challenging and engaging pastime.

In addition to traditional in-person games, poker has seen a significant rise in online platforms, allowing players from around the world to compete against each other. Online poker has introduced new dynamics, such as faster gameplay and the ability to play multiple tables simultaneously, as noted by Thompson (2020). This evolution has expanded the reach of poker, making it more accessible and fostering a global community of players.

Overall, poker's enduring appeal lies in its complexity and the depth of strategy it offers, making it a game that is easy to learn but difficult to master.

2.3 The Related System

To get an idea on developing the system, I've been heavily influenced by the "Air Poker" game in "Usogui" chapters 429-472, as well as the UI design of "Yu-Gi-Oh! Master Duel.".

2.3.1 A brief overview of Air Poker

It'sa 2v2 and two specific games being played at the same time. There are two floors where the top floor has to make hands from the number the bottom players choose but they only get the numbers their specific ally has gotten (so Hal only gets the number Usogui gets). Usogui doesnt really know until later that the cards being dealt is a rule from those above them so keeping that in mind Usogui and Lalo has to depend on those above. Lets say Fukurou is the only person making acutal hands while Hal is just sitting there doing nothing, Usogui will lose definitely. But the top players also depend on the bottom players

The Rules

- Ante: The ante increases by one air-bios every round.
- Bet: The first bet alternates between players after each round. A bet must be done in under thirty seconds. And a raise can only go up to half of the total pot.
- Any air-bios which isn't full due to things like breathing, cannot be used anymore as an ante or a bet. The maximum bet limit is set to the balance of the player with less air-bios.
- Calamity: In the case of a "special way of losing", the loser must release the air of his air-bios, according to the same number of air-bios that he bet that round.
- The definition of who wins and loses: Any player who leaves the chair for more than ten seconds loses. The opponent will be able to use the key to free himself. After the fifth round ends, or in the case where a player cannot aford the ante anymore, the game continues until one player leaves the chair, due to having no more air-bios. In other words, until one player ends up drowning.
- Forbidden conducts: Violence, or unwillingness to give air-bios, not respecting the limited time, or not listen to the referees orders, any behavior judged to be done in order to get in the way of the game will result in an immediate loss.
- Card numbers are totals of a poker hand (A=1, K=13)

- Once bottom players reveal, top players make hands based on number revealed. No physical cards, so all from memory.
- Calamity triggered if both sides used same cards, and both top player fail to make a hand they get 100 secs to do it again.

2.3.2 Yu-Gi-Oh! Master Duel UI

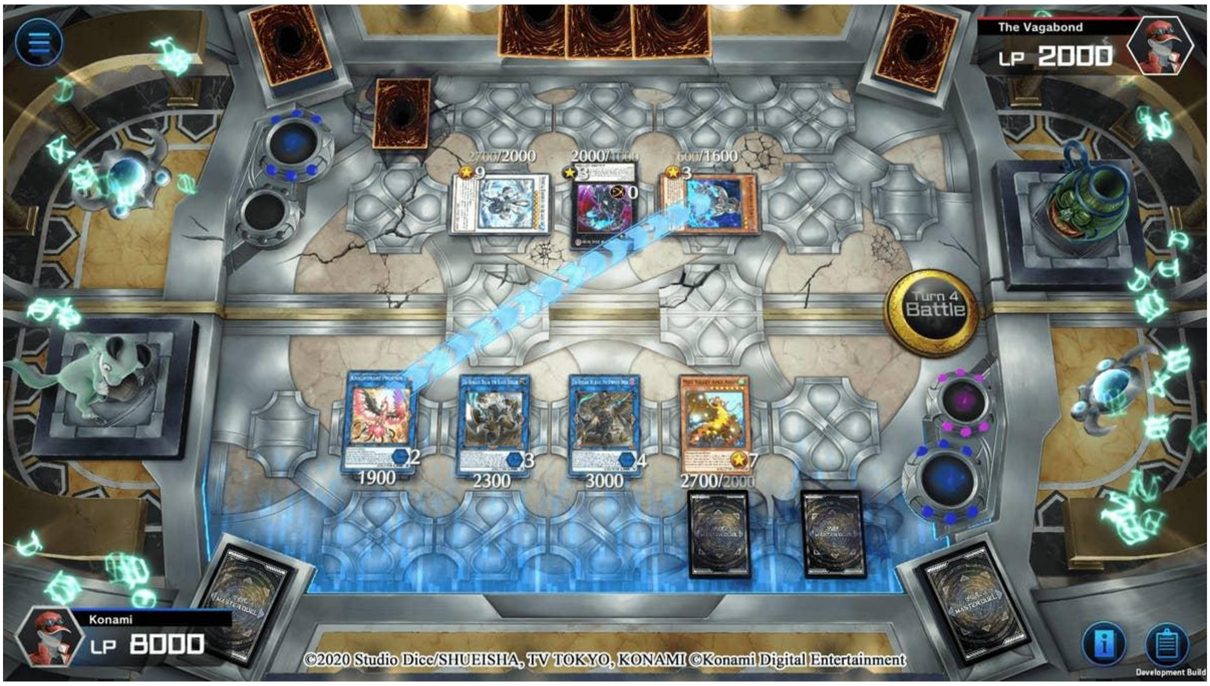


Figure 1 Yu-Gi-Oh! Master Duel UI

2.4 Related Techniques and Methodology

2.4.1 PUN 2

Figure 2 Logo of PUN 2

Photon Unity Networking 2 (PUN 2) is a widely-used networking framework designed to streamline the creation of multiplayer games and applications within Unity. Building on the success of its predecessor, PUN Classic, PUN 2 offers an improved architecture that better supports the demands of modern multiplayer experiences. It integrates seamlessly with Unity, enabling developers to quickly implement real-time, cross-platform multiplayer functionality across various platforms, including PC, mobile, VR, and consoles. PUN 2 provides robust tools for real-time communication, room and matchmaking systems, and the synchronization of game objects, ensuring a consistent game state for all players. Additionally, PUN 2 includes advanced features like interest management, custom serialization, and reliable communication, making it suitable for a wide range of multiplayer game types. Backed by extensive documentation and a strong community, PUN 2 remains a top choice for developers looking to add multiplayer capabilities to their Unity projects.

2.4.2 Firebase

Figure 3 Logo of Firebase

According to Khawas and Shah (2018) Firebase is a relatively new technology for handling large amounts of unstructured data. This paper focuses on the application of Firebase with

Android and aims at familiarizing its concepts, related terminologies, advantages, and limitations. The paper also tries to demonstrate some of the features of Firebase by developing an Android application. They also stated that Firebase provides many functions to the user such as Firebase Cloud Messaging (FCM), Firebase Auth, Real-time Database, and Firebase Storage. Moreover, in term of data storage format, Firebase uses JSON Tree while Relational Database Management System (RDBMS) use tables. Therefore, in terms of efficiency Firebase provides a quick and fast database compared to RDBMS. It is very easy to use the firebase functionality in the Android application. A few lines of codes are all that are needed to link the application with Firebase. In this research paper, it illustrates how to use the basic functions of Firebase in creating Android System. In conclusion, Firebase helps in increasing the efficiency of an Android application as it does not require third party language to communicate with the database.

2.4.3 Unity

Figure 4 Logo of Unity

Unity is a powerful and versatile game development platform used by developers around the world to create both 2D and 3D games, as well as interactive experiences, simulations, and virtual reality (VR) applications. Known for its user-friendly interface and extensive feature set, Unity supports development across multiple platforms, including PC, mobile devices, consoles, VR, and augmented reality (AR).

One of Unity's standout features is its cross-platform capabilities, allowing developers to build a game once and deploy it across various platforms with minimal adjustments. Unity also offers a robust asset store, where developers can access a vast library of pre-made assets, scripts, and tools to accelerate development.

Unity’s scripting is primarily done using C#, and it provides a powerful editor that allows for real-time previews, intuitive scene creation, and comprehensive debugging tools. The platform also supports multiplayer functionality through various networking frameworks, including Photon Unity Networking (PUN), enabling developers to create both local and online multiplayer games.

With a large community, extensive documentation, and ongoing updates, Unity remains a go-to choice for both indie developers and large studios looking to create high-quality, engaging experiences.

2.4.4 Unreal Engine

Figure 5 Logo of Unreal engine

Unreal Engine is a highly advanced game development platform created by Epic Games, known for its powerful graphics capabilities and versatility across a wide range of applications. Originally designed for first-person shooters, Unreal Engine has evolved to become one of the most popular and robust engines for creating not only games but also simulations, films, architectural visualizations, and virtual reality (VR) experiences.

One of the key strengths of Unreal Engine is its visual fidelity. It provides cutting-edge rendering technology, enabling developers to create photorealistic graphics with dynamic lighting, realistic physics, and complex visual effects. This makes it particularly popular in industries that require high-end visuals, such as AAA game development and virtual production for films.

Unreal Engine also features a visual scripting system called Blueprints, which allows developers to create gameplay mechanics, AI behaviors, and other functionalities without needing to write code. This makes it accessible to both programmers and designers, fostering a collaborative development environment. For those who prefer coding, Unreal Engine supports C++, offering deep customization and control over every aspect of the game.

The engine is cross-platform, allowing developers to deploy their projects on PC, consoles, mobile devices, and VR platforms. It also includes powerful tools for networking, animation, audio, and physics, making it a comprehensive solution for game development.

With its extensive documentation, active community, and continuous updates from Epic Games, Unreal Engine is widely used by both indie developers and large studios alike. It has been the backbone for many successful and visually stunning games, as well as other interactive experiences across various industries.

2.4.5 Godot

Figure 6 Logo of Godot

Godot is an open-source game engine that has gained popularity for its flexibility, ease of use, and strong support for 2D and 3D game development. Created by Juan Linietsky and Ariel Manzur, and maintained by a growing community, Godot is a lightweight yet powerful tool that is especially favored by indie developers and small teams.

One of the standout features of Godot is its scene system, which allows developers to structure their projects in a highly modular and hierarchical way. In Godot, everything is a scene, and scenes can be composed of other scenes, enabling easy reuse and management of game elements. This system makes it straightforward to build and organize complex game projects.

Godot supports multiple programming languages, including its own scripting language called GDScript, which is similar to Python and is known for its simplicity and ease of learning. In addition to GDScript, developers can also use C#, C++, and VisualScript, a visual programming language, making Godot accessible to both beginners and experienced programmers.

The engine is well-suited for 2D game development, offering a dedicated 2D engine that is independent of its 3D capabilities. This means that 2D games can be created without the overhead of 3D features, resulting in better performance and more straightforward development. However, Godot's 3D engine has also been significantly improved, providing tools for rendering, physics, animation, and more, allowing for the creation of complex 3D games.

Godot is completely free and open-source, released under the MIT license, which means developers can use and modify the engine without any royalties or restrictions. The engine also has an active community, contributing to its development, creating plugins, and offering support through forums and documentation.

Overall, Godot is a highly flexible and accessible game engine that offers a strong set of features for both 2D and 3D game development, making it a popular choice among indie developers, educators, and hobbyists.

2.4.6 AWS Amplify

Figure 7 Logo of AWS Amplify

AWS Amplify is a development platform by Amazon Web Services that provides a set of tools and services for building scalable mobile and web applications. It offers features similar to Firebase, such as authentication, real-time data, analytics, and cloud storage. Amplify supports a variety of frontend frameworks and integrates with other AWS services for backend functionalities.

2.4.7 Backendless

Figure 8 Logo of Backendless

Backendless is a mobile and web app development platform offering a range of backend services such as user authentication, real-time databases, cloud storage, and push notifications. It provides a visual app development interface and supports server-side logic with custom code, making it a versatile alternative to Firebase.

2.4.8 Supabase

Figure 9 Logo of Supabase

Supabase is an open-source alternative to Firebase that provides a real-time database, authentication, and storage solutions. It uses PostgreSQL as its database engine and offers a set of APIs and tools for building applications. Supabase aims to offer a similar feature set to Firebase but with an emphasis on open-source principles and SQL databases.

2.4.9 Parse Platform

Figure 10 Logo of Parse Platform

Parse Platform is an open-source backend-as-a-service framework that provides cloud-based data storage, user authentication, push notifications, and more. Originally developed by Facebook, it has been maintained by the community and is a viable alternative for developers looking for a self-hosted or open-source solution.

2.4.10 Appwrite

Figure 11 Logo of Appwrite

Appwrite is an open-source backend server designed for web, mobile, and Flutter developers. It provides features such as user authentication, databases, file storage, and real-time capabilities. Appwrite aims to simplify backend development with a focus on ease of use and flexibility.

2.5 Summary

This chapter reviews existing research and systems relevant to the thesis topic. It aims to establish a conceptual framework by evaluating various models and case studies, which will inform the development of the proposed system. The review involves analyzing research articles and current systems to support and guide the system's development.

This next chapter outlines the software development approach used for the project. It emphasizes the importance of selecting an appropriate method to ensure a smooth development process. The chapter focuses on the chosen Software Development Life Cycle (SDLC) model, detailing the stages involved in creating and managing software. It also discusses the software and hardware requirements necessary for the project, highlighting how the SDLC framework helps in guiding development and enhancing code quality and production efficiency.

CHAPTER 3

3 PROJECT METHODOLOGY

3.1 Introduction

This chapter explains the software development approach that is implemented for this system. The appropriate method should be carefully chosen to avoid any difficulties in the development process. We will concentrate on the type of lifecycle of software development used in this project and the software and hardware requirements needed for the project.

Software Development Life Cycle (SDLC) is a framework that describes the tasks performed at each stage of the software development process. It consists of a detailed plan outlining how to create, manage and maintaining software. SDLC help in guiding the development team to develop a system and to improve the quality of the code and the overall phase of production.

3.2 Development Lifecycle Methodology Used in Project

The agile method was chosen as it promotes continuous iteration of development and testing throughout the software development lifecycle of the project. Moreover, agile methodology is flexible and changes can be made in each development phase. To ensure that this application can be used long-term, constant feedback from the user is really important to enhance it is efficiency. Therefore, agile methodology is having been chosen for this project.

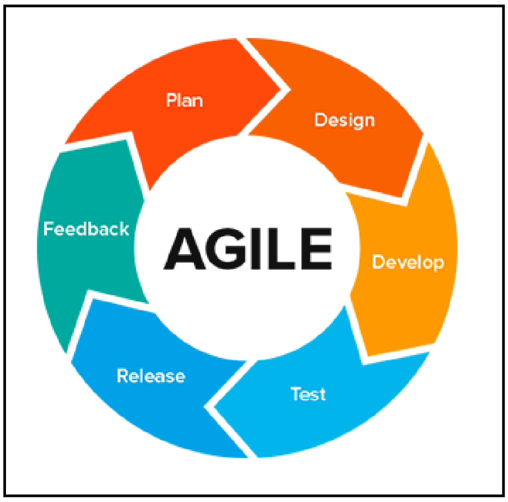


Figure 12 Agile Methodology use for the research

Table 2 Summary of Agile Methodology Justification

Phase	**Description**	**Outcome**
Plan	• Identify target audience and their preferences through surveys and research. • Define functional and non-functional requirements. • Determine necessary hardware and software for development and deployment.	Reference the game 'Water Poker' from 'The Lie Eater'
Design	• Translate requirements into design documents. • Create diagrams including use case, activity, context, data flow, and entity-relationship diagrams, as well as storyboards.	The diagrams that are being used in this project are: 1. Use case diagram 2. Activity diagram 3. Context diagram 4. Data flow diagram 5. Entity-relationship diagram 6. Storyboard
Develop	The language used to create this project is C-Sharp(C#) and the software used to develop this project is Unity and Visual Studio Code. Therefore, only computer users can download this application.	
Test	Several testings is carried out to ensure it fulfil the requirements of the system. This include: 1. Unit testing 2. Integration testing 3. System testing 4. User acceptance testing	• Test Plan • Test cases table

Release	• Deploy the game and gather user feedback. • Perform iterations based on feedback to enhance the game.	
Feedback	• Based on user acceptance testing • Analyse the drawback of the system internally and externally.	

The table provides a summary of the Agile methodology used in a project, outlining the key phases, their descriptions, and expected outcomes.

- Plan Phase: This phase focuses on understanding the target audience and their preferences through surveys and research. It involves defining both functional and non-functional requirements, as well as identifying the necessary hardware and software for development and deployment. The outcome of this phase is a reference to the game "Water Poker" from "The Lie Eater," which informs the project's requirements.

- Design Phase: In this phase, the requirements from the planning stage are translated into detailed design documents. This involves creating various diagrams, including use case, activity, context, data flow, and entity-relationship diagrams, along with storyboards. The outcome is the creation of these diagrams, which provide a visual representation of the system's structure and flow.

- Develop Phase: During this phase, the project is developed using the C# programming language and software tools like Unity and Visual Studio Code. The outcome is a system that can be downloaded and used only on computers, as these tools are geared towards desktop applications.

- Test Phase: This phase involves rigorous testing to ensure the system meets its requirements. The testing process includes unit testing, integration testing, system testing, and user acceptance testing. The outcomes are a test plan and a table of test cases that guide the testing process and ensure the system functions as expected.

- Release Phase: In this phase, the game is deployed to users, and feedback is collected. Based on this feedback, iterations are performed to enhance the game. The outcome is an improved version of the game based on real user input.

- Feedback Phase: The final phase focuses on analyzing the results of user acceptance testing and identifying any internal and external drawbacks of the system. This analysis informs future improvements and refinements. The outcome is a thorough understanding of the system's strengths and weaknesses.

Overall, the table outlines a structured approach to project development using Agile principles, emphasizing continuous feedback and iterative improvement.

3.3 Justification of Project Methodology

Choosing the Agile methodology for a project comes with several justifications, particularly for projects that are complex, dynamic, or involve significant user interaction and feedback. Here are some reasons why Agile might be the best choice for your project:

- Flexibility and Adaptability: Agile methodology is known for its flexibility, allowing teams to adapt to changes quickly. In a project where requirements may evolve or where new insights might emerge during development, Agile provides the ability to incorporate these changes without disrupting the entire project timeline.

- Continuous User Feedback: Agile emphasizes regular interaction with stakeholders and end-users through iterative cycles. This continuous feedback loop ensures that the project stays aligned with user needs and expectations, reducing the risk of delivering a product that doesn’t meet user requirements.

- Incremental Development: With Agile, the project is broken down into smaller, manageable parts, known as iterations or sprints. Each iteration delivers a functional part of the project, which can be tested and evaluated. This approach ensures that progress is visible and tangible, providing a sense of accomplishment and allowing for early detection and correction of issues.

- Risk Management: Agile's iterative nature allows for frequent testing and evaluation, which helps identify risks early in the development process. By addressing these risks incrementally, the likelihood of major issues arising at the end of the project is minimized.

- Improved Collaboration: Agile promotes collaboration among team members and stakeholders through regular meetings (e.g., daily stand-ups, sprint reviews). This ongoing communication ensures that everyone is on the same page, fostering a collaborative environment where issues can be resolved quickly and effectively.

- Focus on Quality: Agile encourages regular testing and review at the end of each sprint, which ensures that quality is maintained throughout the project. By the time the final product is delivered, it has undergone multiple rounds of testing and refinement, resulting in a higher-quality outcome.

- User-Centric Approach: Agile is particularly well-suited for projects where user experience is critical. The methodology's focus on user feedback and iterative design helps ensure that the end product is closely aligned with what users want and need, leading to higher satisfaction and better adoption rates.

- Efficiency and Speed: Agile's iterative approach allows for the rapid delivery of functional components of the project. This can be particularly important in fast-moving industries where getting a product to market quickly can provide a competitive advantage.

Given these justifications, Agile is a strong choice for projects that require adaptability, close collaboration with stakeholders, and a focus on delivering high-quality, user-centered outcomes in an efficient and manageable way.

3.4 Summary

Project Methodology outlines the approach used for the software development process, specifically focusing on the Agile methodology. The chapter begins by explaining the importance of choosing an appropriate software development lifecycle (SDLC) to ensure smooth project execution. The Agile methodology was selected for this project because of its flexibility, iterative nature, and emphasis on continuous user feedback, which are crucial for adapting to changes and improving the final product. The chapter also provides a detailed overview of the Agile methodology's phases, including planning, design, development, testing, release, and feedback, each with specific activities and expected outcomes.

Project Analysis and Design will build upon the methodology outlined in Chapter 3 by diving into the specifics of the project’s analysis and design processes. Since Agile emphasizes iterative design and continuous feedback, Chapter 4 will likely detail how the project requirements were analyzed and how the design evolved through various iterations. This chapter is expected to describe the specific diagrams and models created during the design phase, as mentioned in Chapter 3, such as use case diagrams, activity diagrams, and data flow diagrams. These design artifacts will be crucial in translating the project requirements into a functional system architecture.

In essence, while Chapter 3 establishes the framework and rationale behind choosing Agile, Chapter 4 will demonstrate how this methodology was applied to analyze and design the project, ensuring that it meets user needs and functions effectively.

CHAPTER 4

4 PROJECT ANALYSIS AND DESIGN

4.1 Introduction

Requirement analysis is one of the important steps to start developing a system. To get different perspectives from stakeholders, gathering requirements is required to understand the background and problems the respondents faced, and to get their opinions on this system.

4.2 Project Planning

The planning phase is the initial step in the Agile methodology, which sets the foundation for the development of the Mind Poker game. In this phase, stakeholders identify the core problem: the lack of intellectually stimulating online games that focus on enhancing cognitive abilities like intelligence, memory, mathematical skills, and strategic thinking. Current games often prioritize luck over skill and do not challenge the mental faculties of players, which creates a gap in the market for a more skill-based, cognitive game.

To address this gap, the target users for Mind Poker are identified. These users include individuals who are interested in cognitive development, strategic gaming, and those seeking a competitive and skill-based challenge. The game's unique design integrates intellectual challenges with traditional poker mechanics, ensuring that player strategies, rather than luck, drive the outcome of each game. This approach creates an engaging and mentally stimulating experience for players.

During the planning phase, research is conducted on the current state of cognitive and strategy games, as well as studies in cognitive development. This research helps identify the necessary functional and non-functional requirements for the system, ensuring that it meets the needs of its target audience. Additionally, competitive analysis of similar brain-training games and multiplayer strategy games helps refine the game design and development process.

Stakeholder meetings and surveys help further refine the objectives of Mind Poker. The requirements for hardware and software, such as Unity, Visual Studio Code, and database management systems, are defined. These tools are chosen to provide a seamless online multiplayer experience while ensuring that players are consistently challenged by the increasing difficulty of the game. The system's architecture is planned to allow for scalability and flexibility, aligning with Agile's iterative approach to development. This planning process ensures that the game will offer a high degree of playability and cognitive benefits while fostering a competitive and engaging environment.

The planning phase also considers the project's constraints, such as the complexity of integrating multiple databases and the time required to develop programming skills. These challenges are addressed by allocating sufficient time for learning and testing, with an emphasis on minimizing potential errors during the development and implementation phases. By proactively identifying these constraints, the project planning aims to mitigate risks and ensure the successful delivery of Mind Poker as a unique and rewarding intellectual challenge.

4.3 Project Requirements

4.3.1 Gathering and Analysis of Requirements

The gathering and analysis of requirements is a crucial phase in the development of Mind Poker, as it establishes the foundation for the game's functionality, user experience, and overall performance. This process involves collecting input from various stakeholders, conducting research, and performing a detailed analysis to define the essential features and components of the game.

Gathering Requirements

1. Stakeholder Input: The primary stakeholders of Mind Poker include potential players, game designers, and developers. Feedback from these groups is gathered through surveys, interviews, and market research. This information helps identify the needs, preferences, and pain points of the target audience—those who seek intellectually challenging and skill-based games.

2. Competitive Analysis: Research is conducted on existing games in the cognitive development and strategy genre, including brain-training apps and online multiplayer strategy games. The strengths, weaknesses, and unique features of these games are analyzed to ensure that Mind Poker provides a differentiated and compelling experience.

3. User Research: Surveys and questionnaires are distributed to potential users to gather insights into their preferences for gameplay mechanics, cognitive challenges, and multiplayer dynamics. This helps ensure that the game is designed with a user-centered approach, meeting the expectations of players who seek a mentally stimulating and strategic gaming environment.

4.3.2 Hardware and Software Requirements

In the development of Mind Poker, identifying the necessary hardware and software requirements is essential to ensure the project runs smoothly and meets both performance and functionality expectations. The hardware and software chosen for the project will significantly influence the development process and user experience. Below is an explanation of the hardware and software requirements, as detailed in Tables 3 and 4.

Table 3 Details of hardware requirements for the project development

Hardware	Specification	Estimated Budget
Personal Computer	Intel Core i3 2.0 GHz or equivalent or higher 4GB of RAM or higher 320 GB hard drive storage or higher WLAN and Gigabit Ethernet	RM 1500
Smartphone	Android 8.0 and above	RM 500

Table 4 Details software requirement for the development

Software	Purpose	Estimated Budget
Unity	To create and edit	Free
Firebase	To store database	Free
Visual studio code	To edit function code	Free
Microsoft Office Word	To prepare project documentation	Free
Pun 2	To create an online environment	Free

4.3.3 Functional Requirements

- Online multiplayer capabilities, allowing players to compete against others in real-time.
- A dynamic scoring system that evaluates a player's intelligence, memory, and mathematical skills.
- Progressive levels of difficulty that increase as the player choose in the game.
- Strategic gameplay mechanics where players' decisions influence the outcome, minimizing the role of luck.
- Seamless integration of game functions, such as matchmaking, leaderboard rankings, and in-game achievements.

4.3.4 Non-Functional Requirements

- Performance: The game must run efficiently on desktop platforms with minimal lag or delays.
- Security: The system must protect player data and ensure that interactions are secure during online gameplay.
- Scalability: The game architecture should be scalable to support a growing number of users without compromising performance.
- User Interface: The design should be intuitive and engaging, allowing users to easily navigate through game menus and features.

4.4 Summary

Chapter 4 of the Mind Poker development documentation details the critical stages of requirement analysis and project planning. It begins with an introduction to the importance of gathering and analyzing requirements to ensure that the game meets the needs of its target audience—those who seek intellectually stimulating and skill-based games. The project planning phase focuses on identifying the core problem: the lack of cognitive and strategic online games that prioritize skill over luck. This phase involves defining the target users, conducting research on cognitive and strategy games, and refining the project's objectives through stakeholder input.

The chapter also outlines the specific hardware and software requirements necessary for the development of Mind Poker, ensuring that the project runs smoothly and meets performance expectations. The functional requirements detail the essential features, such as online multiplayer capabilities, a dynamic scoring system, and strategic gameplay mechanics, while the non-functional requirements address performance, security, scalability, and user interface considerations.

With the foundational requirements and planning established, Chapter 5, "Project Design," will build upon these insights by detailing the architectural framework and design strategies that will bring Mind Poker to life. This chapter will explore the technical design of the game, including system architecture, database design, user interface layout, and the integration of various game functions. The design phase is crucial as it translates the gathered requirements into a concrete plan for development, ensuring that the game's features, performance, and user experience align with the objectives set out in the planning phase.

CHAPTER 5

5 PROJECT DESIGN

5.1 Introduction

This chapter applies the knowledge of object-oriented analysis and design. Object-oriented analysis is the procedure of identifying software requirements and developing software specifications in terms of the object model. The focus of this step is to combine data and function in one analysis rather than separating it.

5.2 Project Design

In this phase, the business requirements are translated into the design specifications for the development. Several diagrams will be used to guide the developer in creating this system which are use case diagram, data flow diagram and entity relationship diagram. Other than that, a storyboard also will be used to give a picture to the developer on the expected user interface of the system.

5.2.1 Use case Diagram

This Use Case diagram: Lobby Part represents the interactions between a user and the system, specifically focusing on the lobby part of the application. The diagram identifies a single actor, "User," who can perform a variety of actions within the system:

- Register: The user must first register an account to access the system's functionalities.
- Login: After registering, the user can log into the system.
- Logout: The user can log out of the system when they are done using it.
- Change Username: The user has the option to change their username after logging in.
- Start Game: The user can initiate a game session from the lobby.
- Cancel Game: If the user changes their mind, they can cancel a game that has been initiated.
- View Score History: The user can view their past game scores. This action has two extensions:
    1. View Win: Allows the user to view their winning scores.
    2. View Lose: Allows the user to view their losing scores.

This diagram outlines the basic flow of interactions that a user can perform in the lobby part of the application, providing a clear view of the functionalities available to the user.

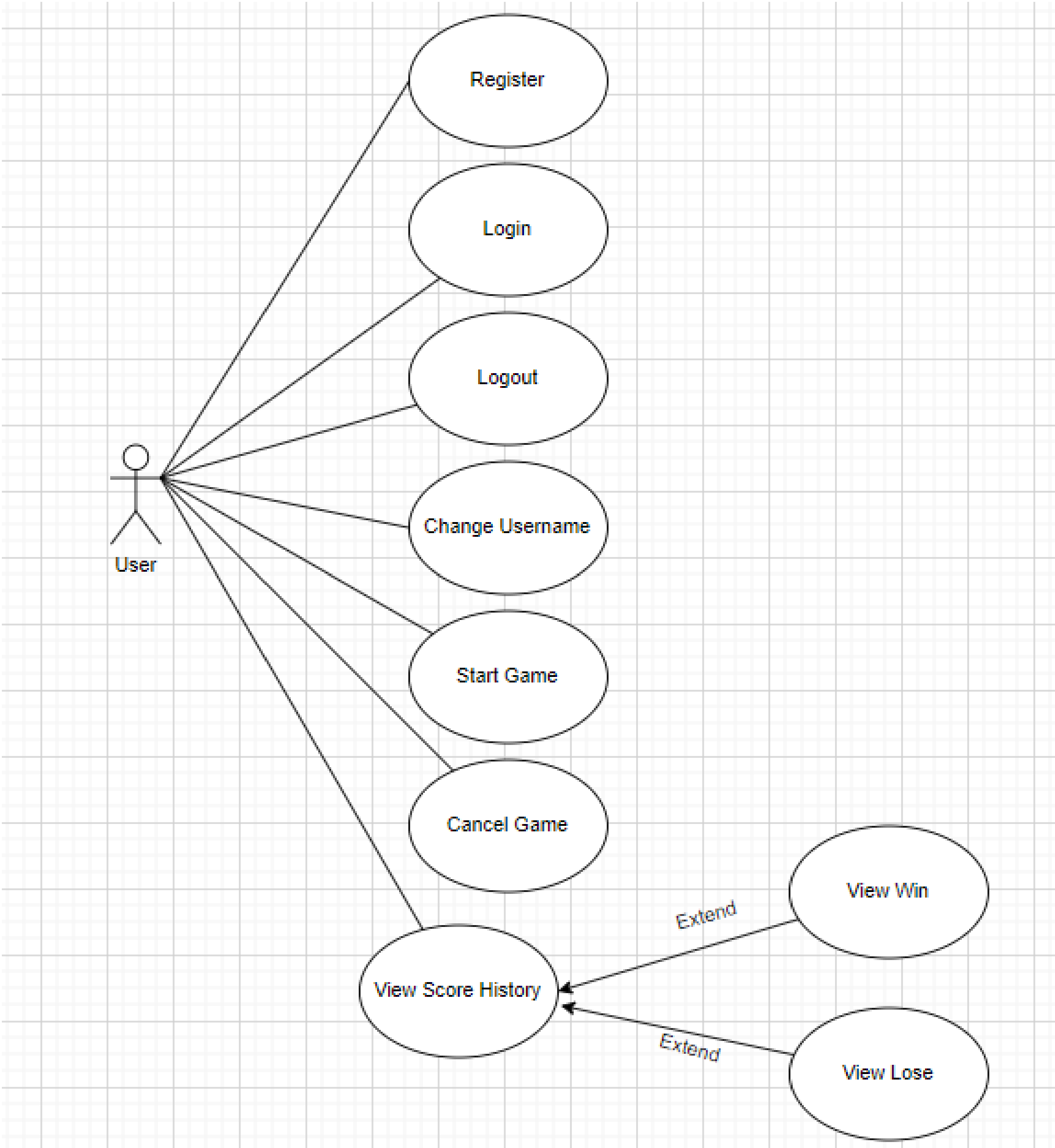


Figure 13 Use Case Diagram: Lobby Part

This Use Case diagram: Game Part represents the actions that a user can perform during the gameplay portion of the system. The user is the only actor in this diagram, and they can engage in various activities related to playing the game:

- Start Game: The user begins the game session.
- Create Game Room: The user can create a new game room where other players can join.
- Join Game Room: The user can join an existing game room to play with other participants.
- Play Against Random Player: The user has the option to play a game against a randomly selected player.
- View Number Card: The user can view the number cards they possess.
- View Heart: The user can view the hearts they have.
- View Phase: The user can see the current phase of the game.
- View Phase Time: The user can see the time allocated for the current phase.
- Choose Number Card: The user selects a number card from their hand to play.
    - Get Number Card Back: An extension of the "Choose Number Card" action, allowing the user to retrieve the number card if needed.
- Choose Poker Card: The user selects a poker card from their hand to play.
- View Opponent Number Card: The user can view the number cards played by the opponent.
- View Opponent Poker Card: The user can view the poker cards played by the opponent.
- View Opponent Heart: The user can see the hearts that the opponent has.

This diagram outlines the sequence of interactions that a user can perform within a game, highlighting the key actions available during gameplay.

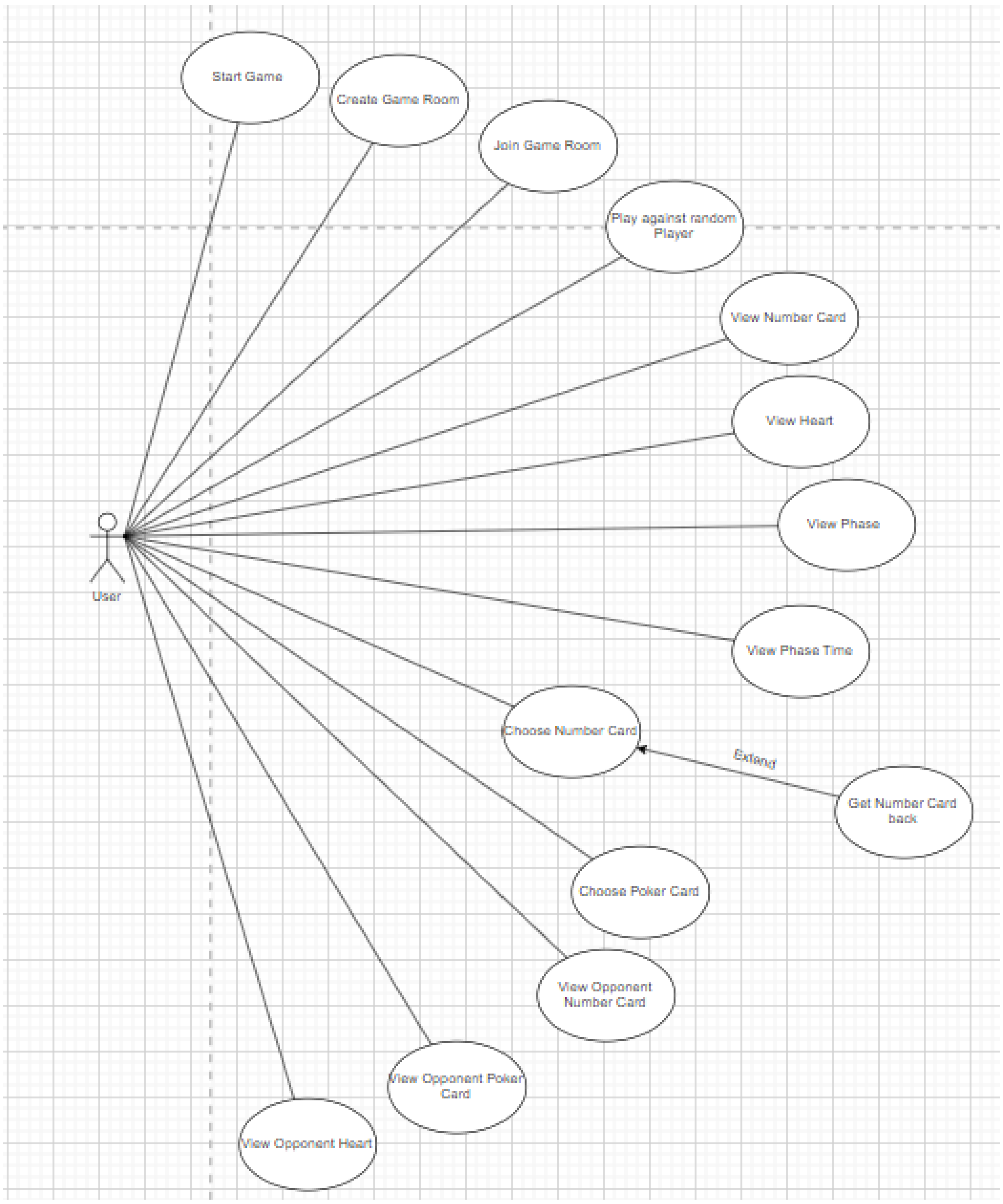


Figure 14 Use Case Diagram: Game Part

5.2.2 Interface Design – Storyboard

This system consists of four pages. The purpose of making a storyboard is to plan and act as a guideline in making the application. A storyboard is vital as it conveys how each of the pages of the application communicates with each other and to know the flow of an application.

a) Login Page

The application will start with the sign-in page. The user is required to input their email and password into the corresponding fields. Upon entering these details, the user can click the "Login" button to authenticate and access the system. If the user does not have an account, they can click the "Register" button, which will redirect them to the registration page. There is also a "Feedback Text" area on the right side of the page, where the system may display messages such as login errors, successful login, or other important notifications to the user. This provides immediate feedback based on the user's actions on this page.

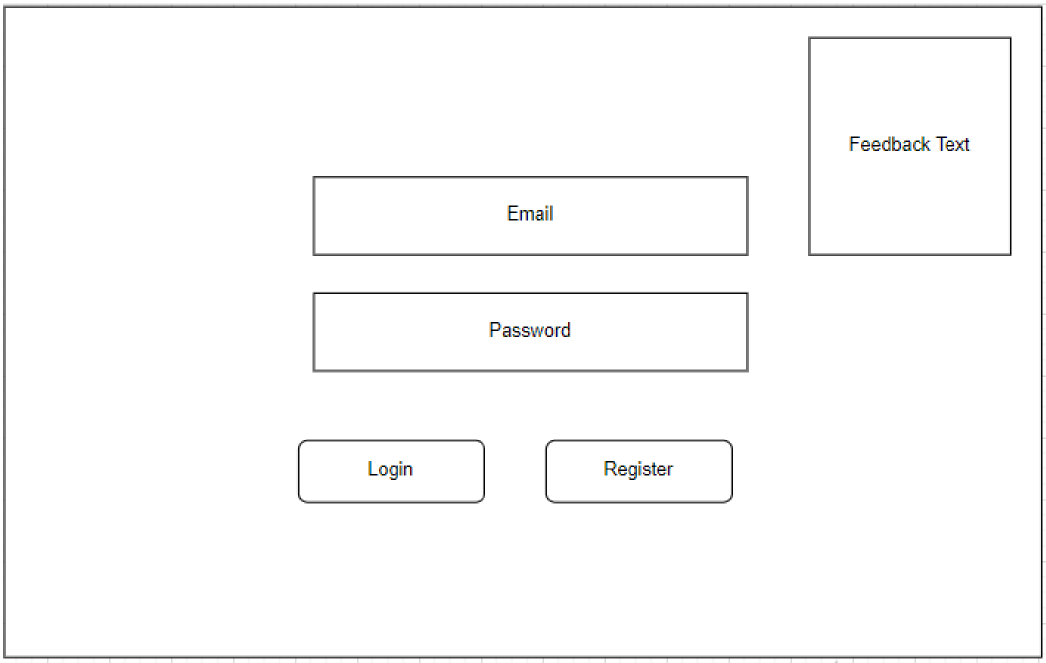


Figure 15 Login Page Storyboard

b) Lobby Page

The lobby page of the application is designed to allow the user to manage their profile and prepare for the next game session. At the top right corner, the user can see their current username or unique identifier (UID). If the user wishes to change their username, they can click the "Change Username" button on the left side, which reveals an input field "Change Username Text" and a "Confirm" button. The user can then type in a new username and confirm the change. Additionally, there is a "Logout" button on the left side that allows the user to log out of the system, returning them to the login page. In the center of the page, the "Play" button is prominently displayed, enabling the user to start a new game session and go to the Loading Page. At the bottom left, there is a "Win/Lose Text" area that displays the outcome of the user's previous game, indicating whether they won or lost. This setup ensures that the user has all the necessary options at their fingertips to manage their profile and seamlessly transition into gameplay.

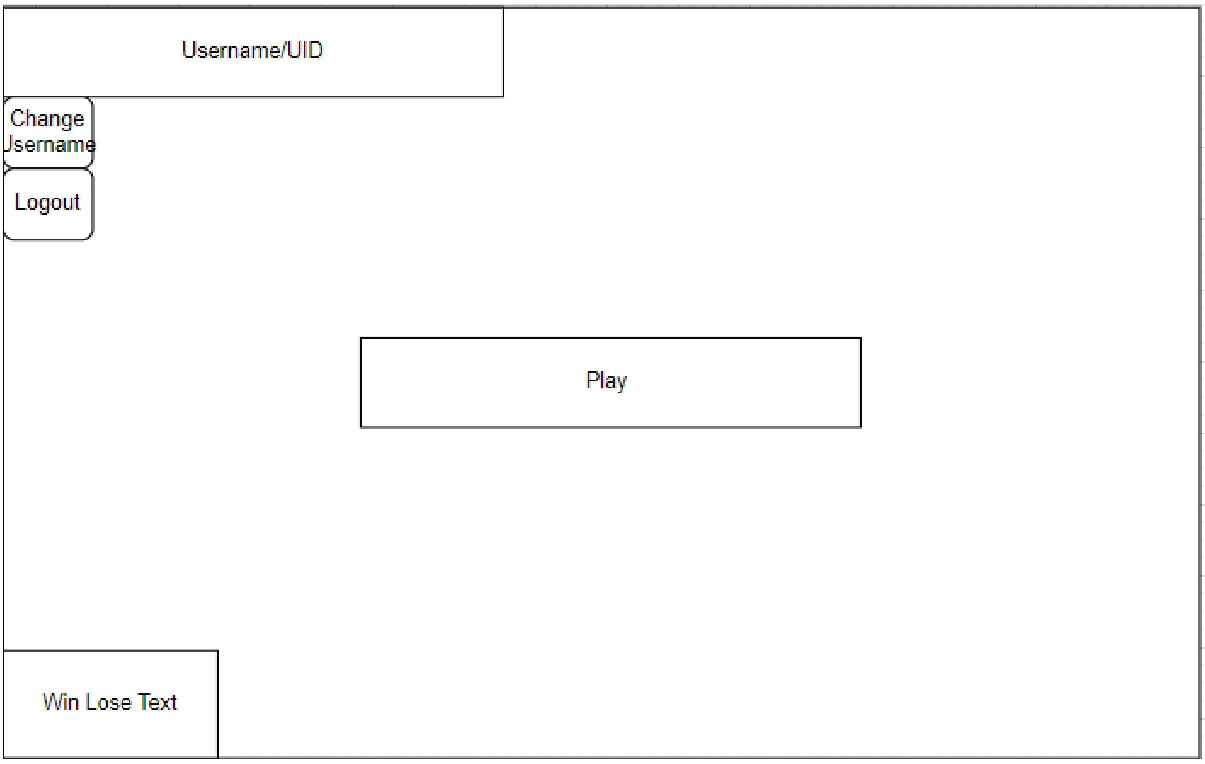


Figure 16 Lobby Page Storyboard

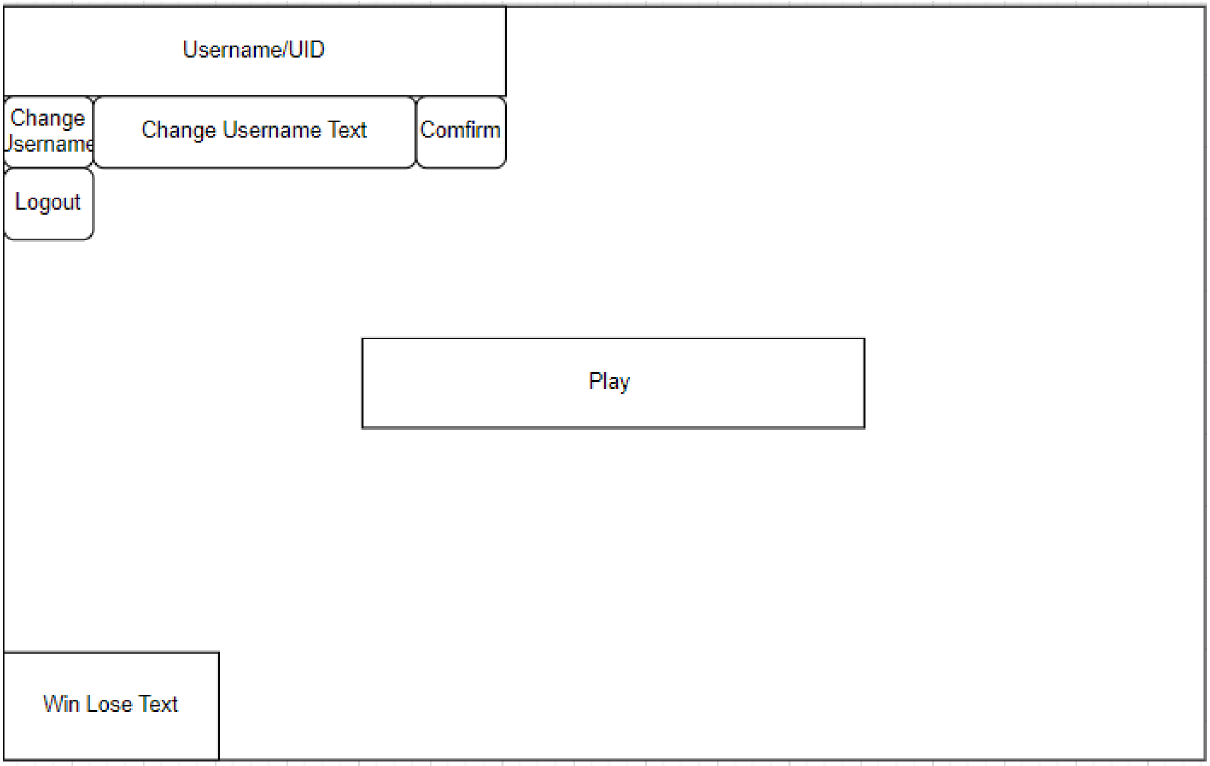


Figure 17 Lobby Page after press Change Username Button Storyboard

c) Loading Page

This loading page is where the player waits for another player to join and match up for a game. At the center of the page, there is a "Waiting For Opponents..." message indicating that the system is searching for another player. Beneath this message, a "Cancel" button is provided, allowing the player to exit the waiting state and return to the previous screen if they decide not to continue waiting. This page is essential for transitioning between the lobby and the game, ensuring that both players are ready before the game begins.

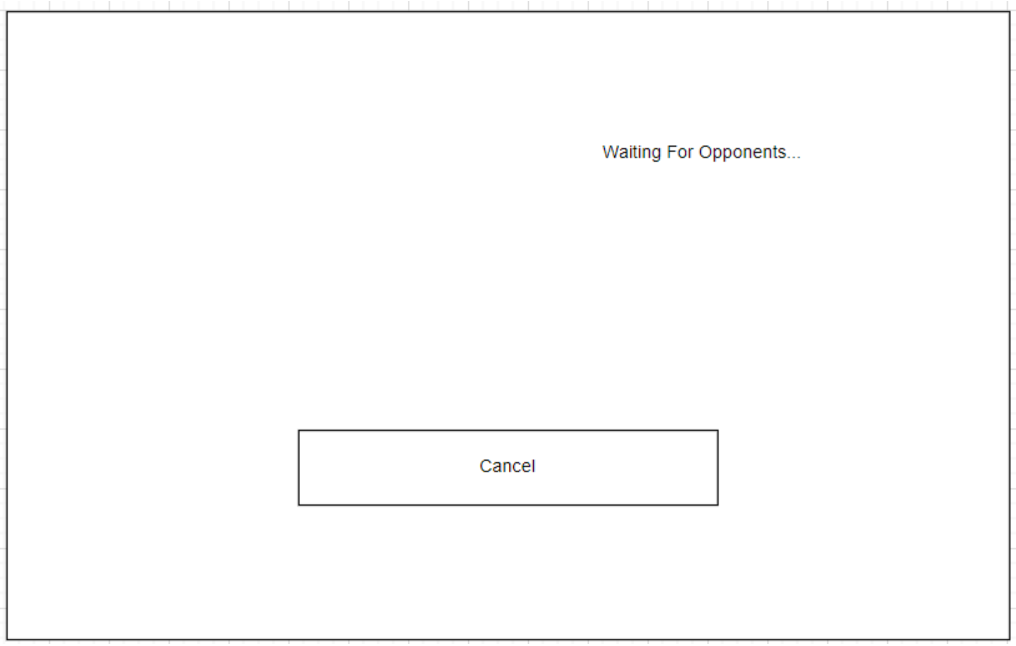


Figure 18 Loading Page Storyboard

d) Game Page

The game page is the central interface where players interact during the card game. It is designed with clarity in mind, allowing players to engage seamlessly with various game elements. Players can select a card from their hand and place it in the designated Player Number Card Slot. Once a card is placed, the Confirm button finalizes its placement. If a player changes their mind, the Down button allows them to remove the card from the slot and return it to their hand. The Phase Text and Phase Time display the current game phase and the remaining time, helping players stay informed and manage their moves efficiently.

Players can view the opponent's card slots, though the actual cards in the Opponent Number Card slots remain hidden to maintain strategic gameplay. The Pick Poker button, when clicked, reveals a panel displaying all available poker cards, allowing the player to choose a card and place it in their respective slot. Heart icons on the interface represent the lives or health of both the player and the opponent, visualizing the competitive aspect of the game. This well-organized interface ensures that players can make strategic decisions and fully engage in the gameplay.

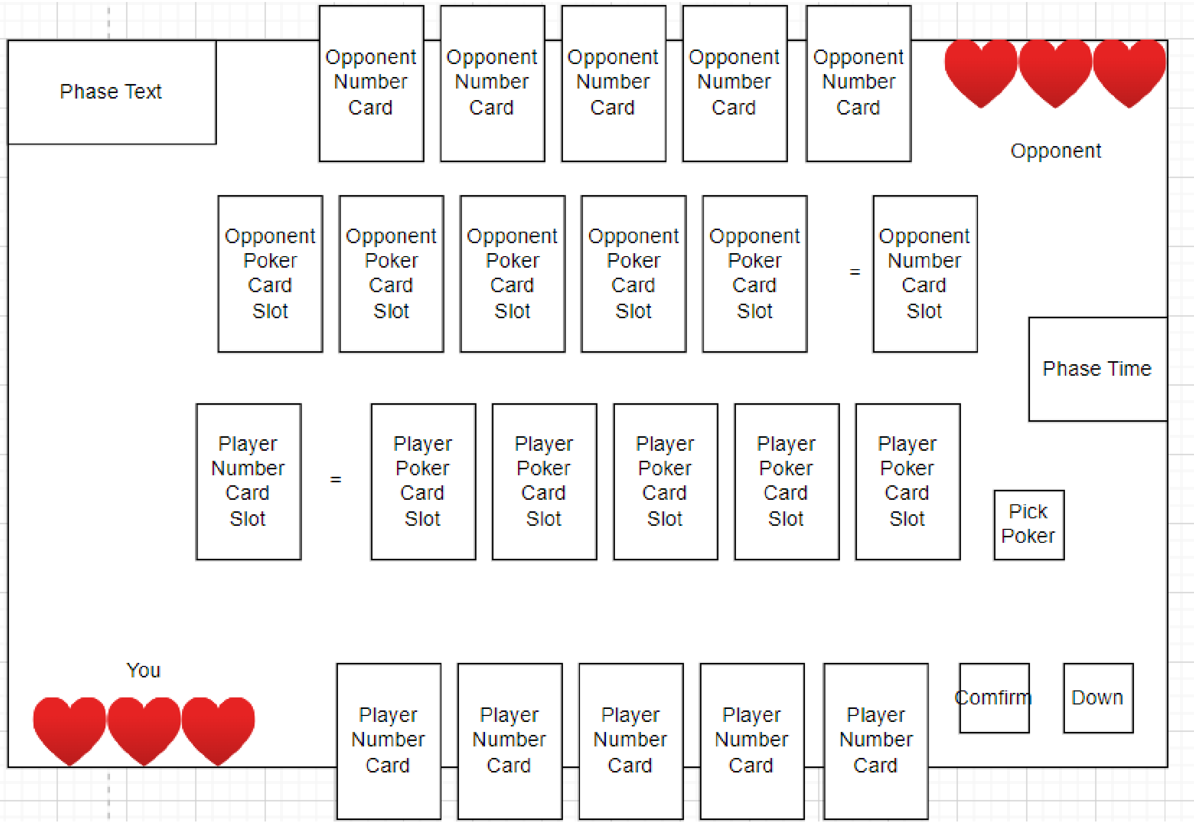


Figure 19 Game Page Storyboard

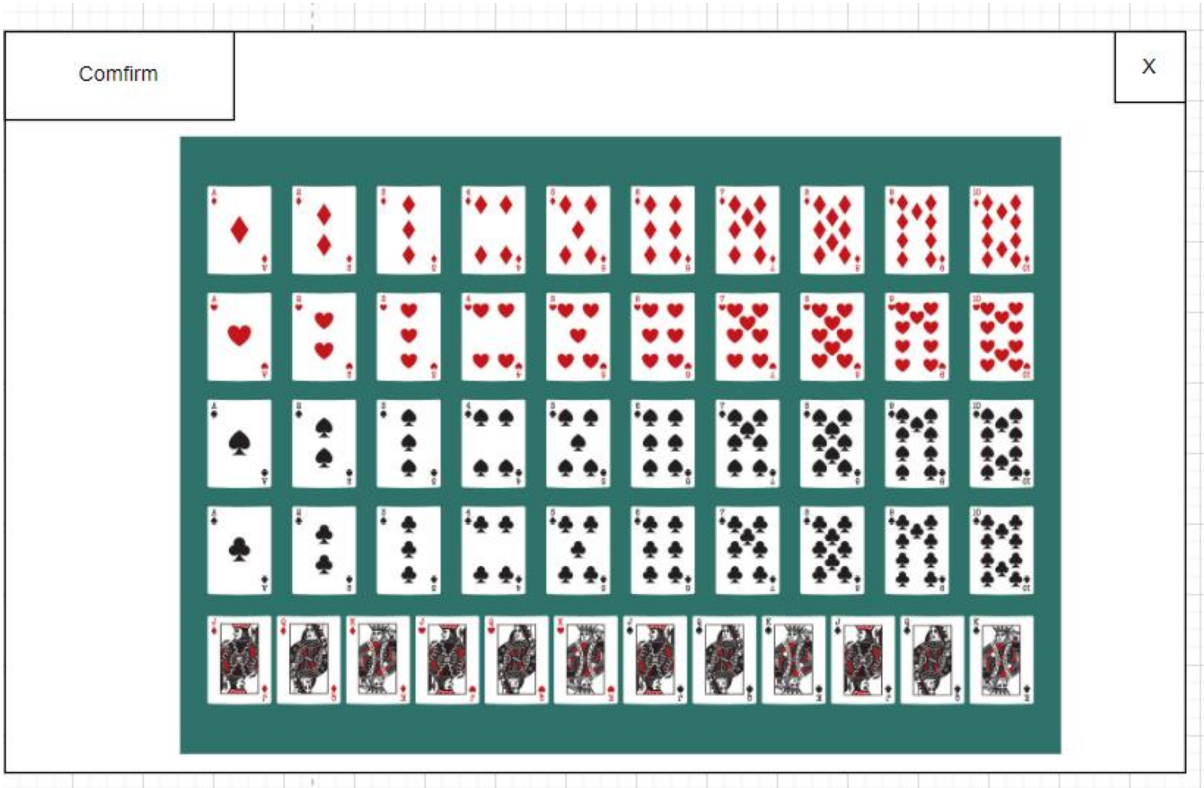
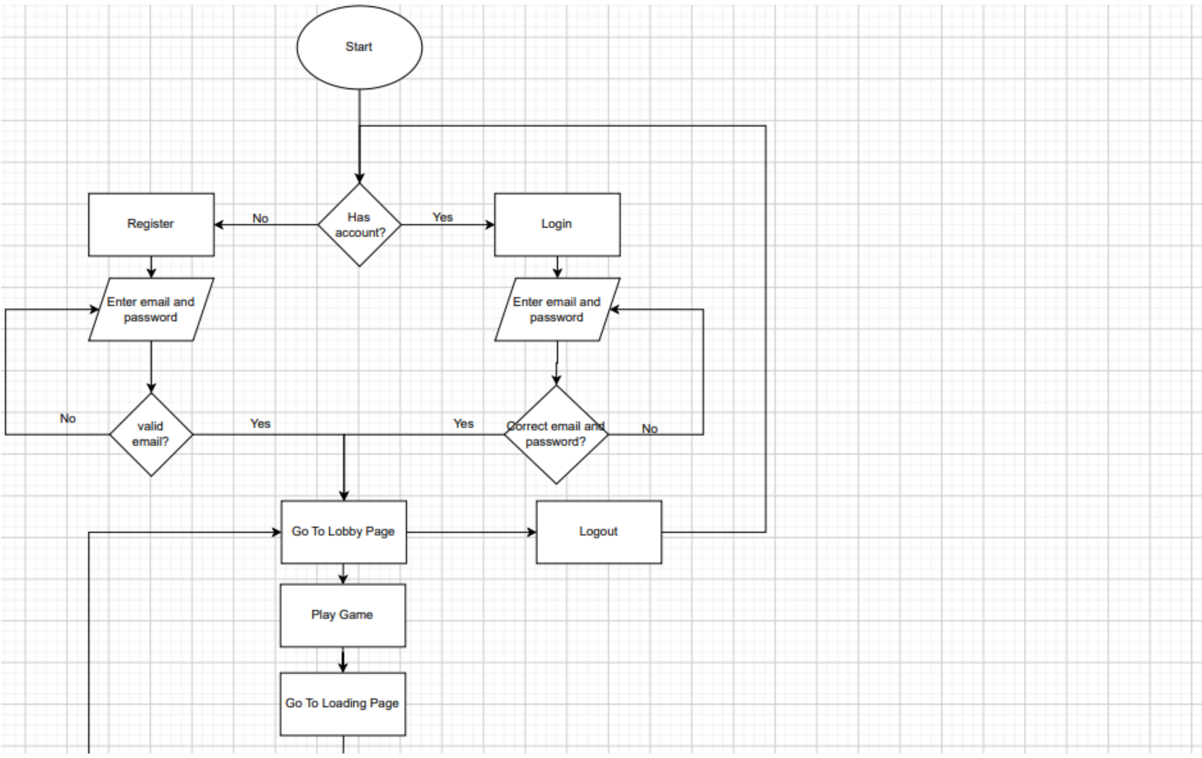
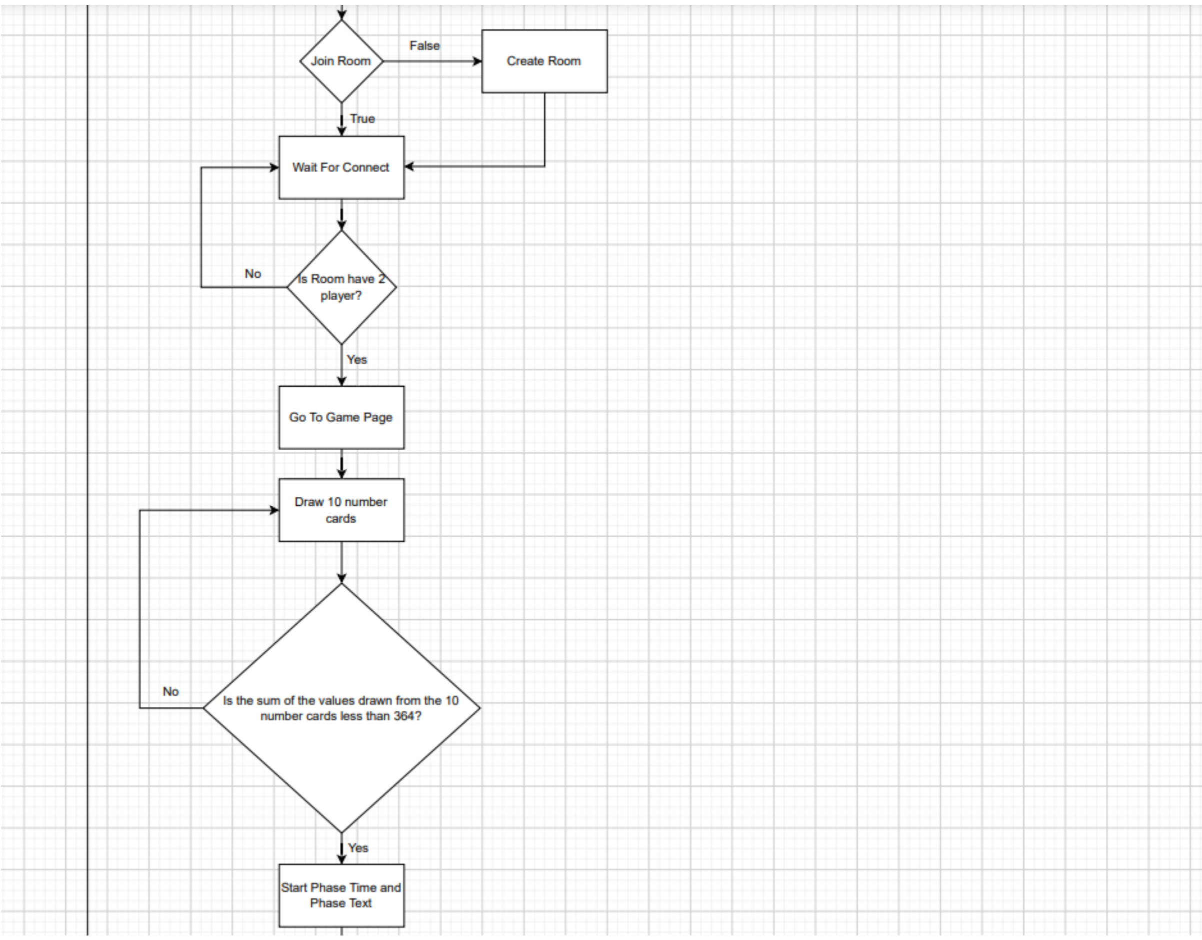
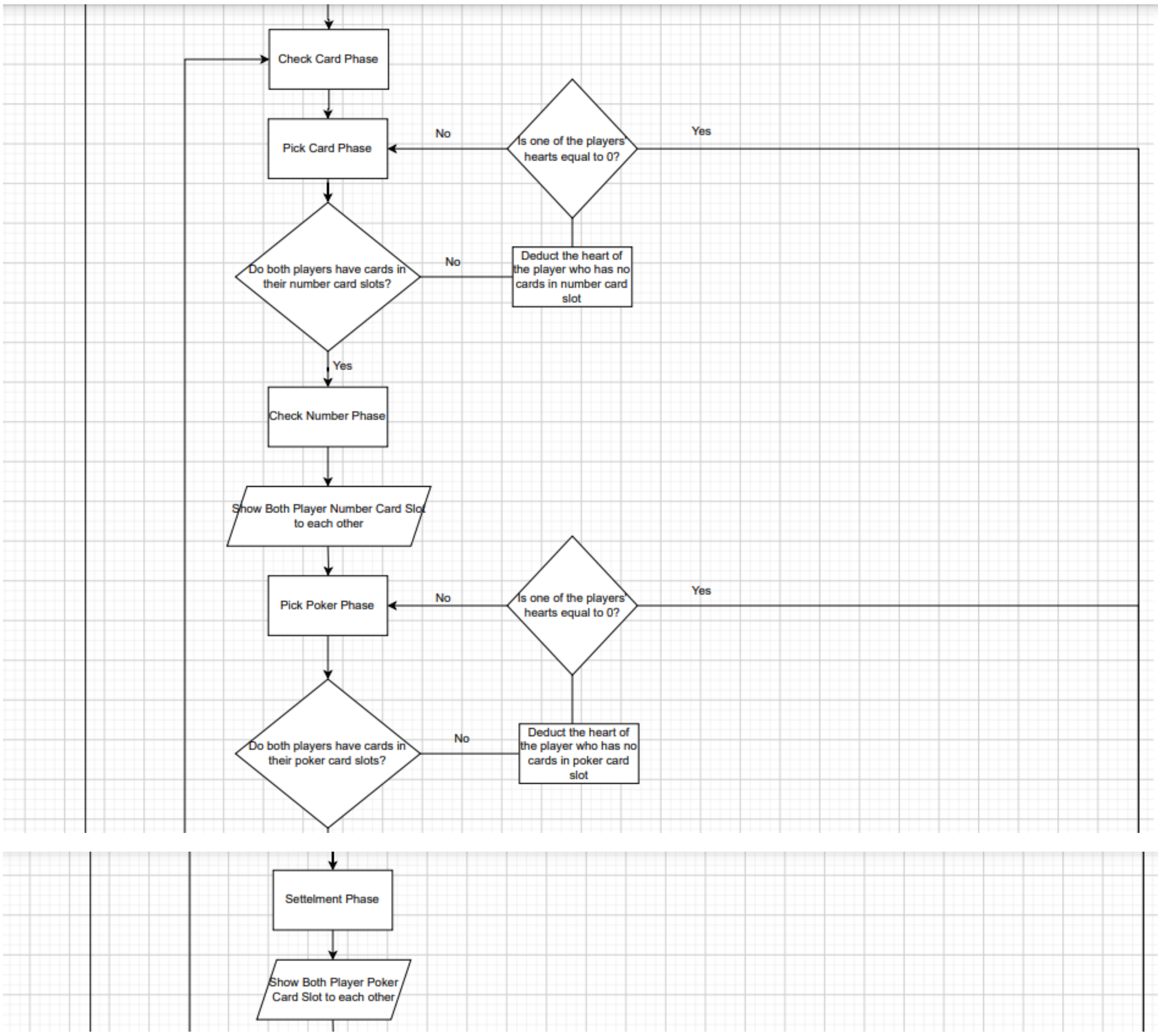


Figure 20 Poker Panel Storyboard

5.2.3 Flowchart






Check Card Phase
Pick Card Phase
No
Is one of the players' hearts equal to 0?
Yes
Do both players have cards in their number card slots?
No
Deduct the heart of the player who has no cards in number card slot
Yes
Check Number Phase
Show Both Player Number Card Slot to each other
Pick Poker Phase
No
Is one of the players' hearts equal to 0?
Yes
Do both players have cards in their poker card slots?
No
Deduct the heart of the player who has no cards in poker card slot
Settelment Phase
Show Both Player Poker Card Slot to each other

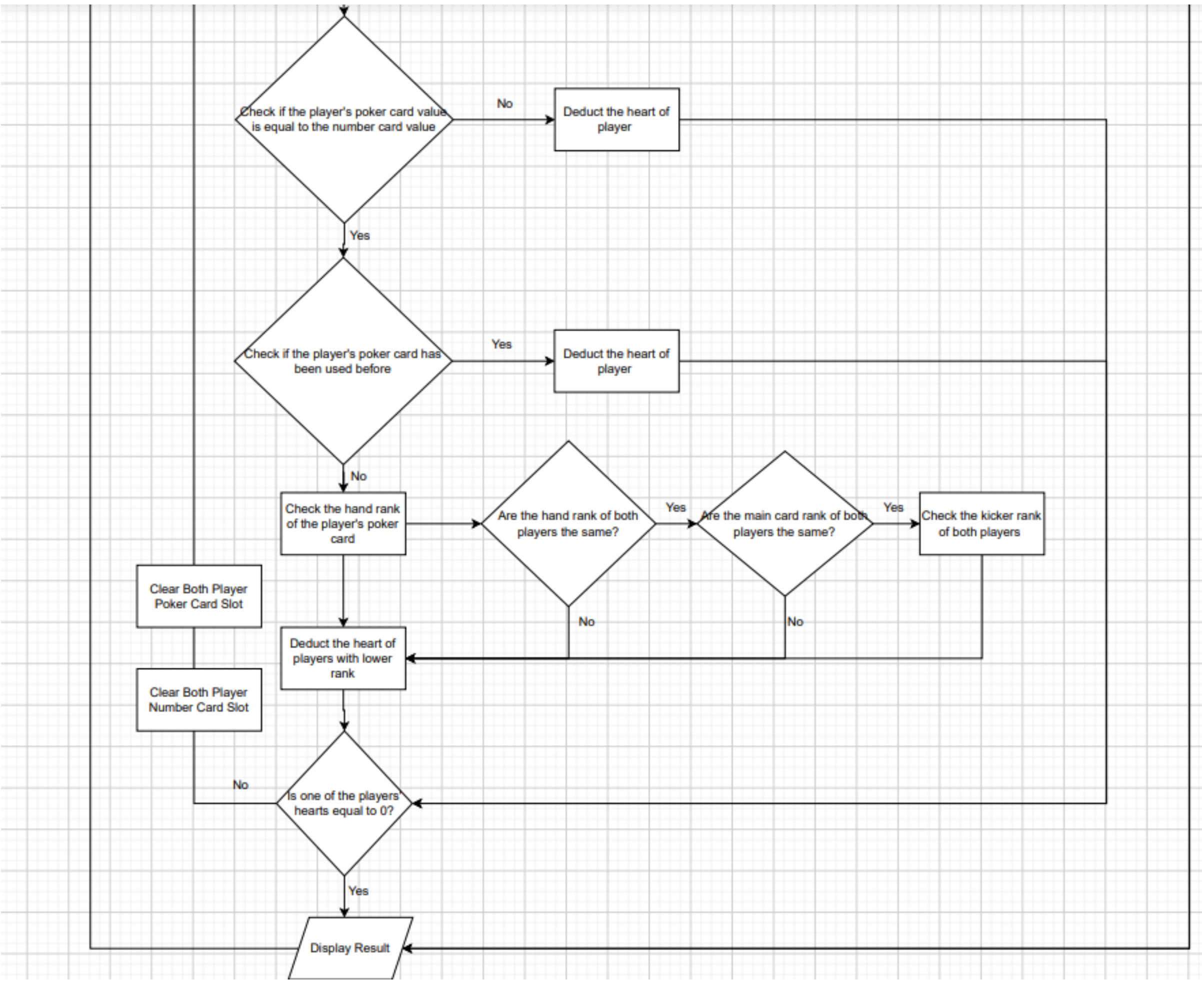


Figure 21 Flowchart

5.2.4 Entity Relationship Diagram

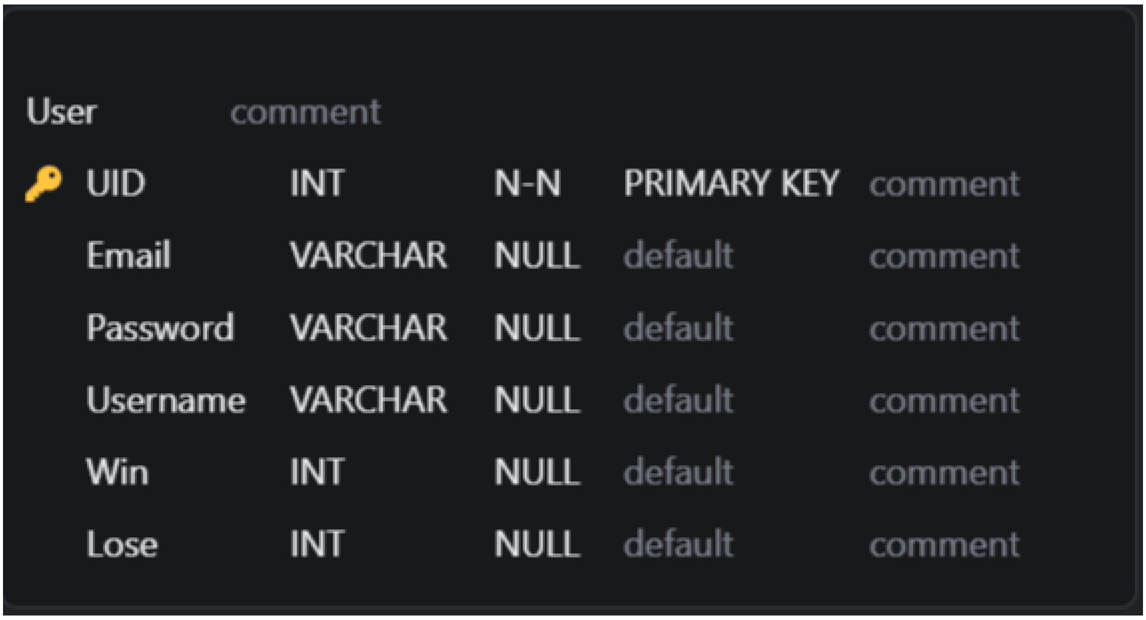


Figure 22 ERD

```
CREATE TABLE User
(
  UID      INT      NOT NULL DEFAULT PRIMARY KEY,
  Email    VARCHAR NULL    ,
  Password VARCHAR NULL    ,
  Username VARCHAR NULL    ,
  Win      INT      NULL    ,
  Lose     INT      NULL    ,
  PRIMARY KEY (UID)
);
```

```
type User {
  uid: ID!
  email: String
  password: String
  username: String
  win: Int
  lose: Int
}
```

Table 5 ERD code

5.3 Summary

The chapter sets the foundation for the next phase, which is "Project Development." By establishing a thorough object-oriented analysis and design process, the groundwork is laid for the actual development of the system. The design specifications and diagrams created in this chapter will serve as essential references for developers as they begin to build the system in Project Development. This ensures that the development phase is aligned with the planned architecture and user interface, leading to a more efficient and organized development process.

CHAPTER 6

6 PROJECT DEVELOPMENT

6.1 Project Development Process

In this phase, the developer translates the design into the programming language. The language used in this system is C# and the program that will be used is Unity and Visual Studio Code. This project uses Firebase services for the online cloud platform. Firebase provides a real-time database that can synchronize the data across clients. Other than that, Firebase also provides storage to store data. This prohect also uses PUN 2 to achieve real time environment. PUN 2 provides a real time server to synchronize the data across clients. The project implements C#, PUN 2 and Firebase Database. The project will be capable of running on any CPU or laptop.

6.2 User Manual

6.2.1 Login & Register Activity

Figure 23 show the login and register activity. The Login page of Mind Poker require the user to login or register by using email and password.

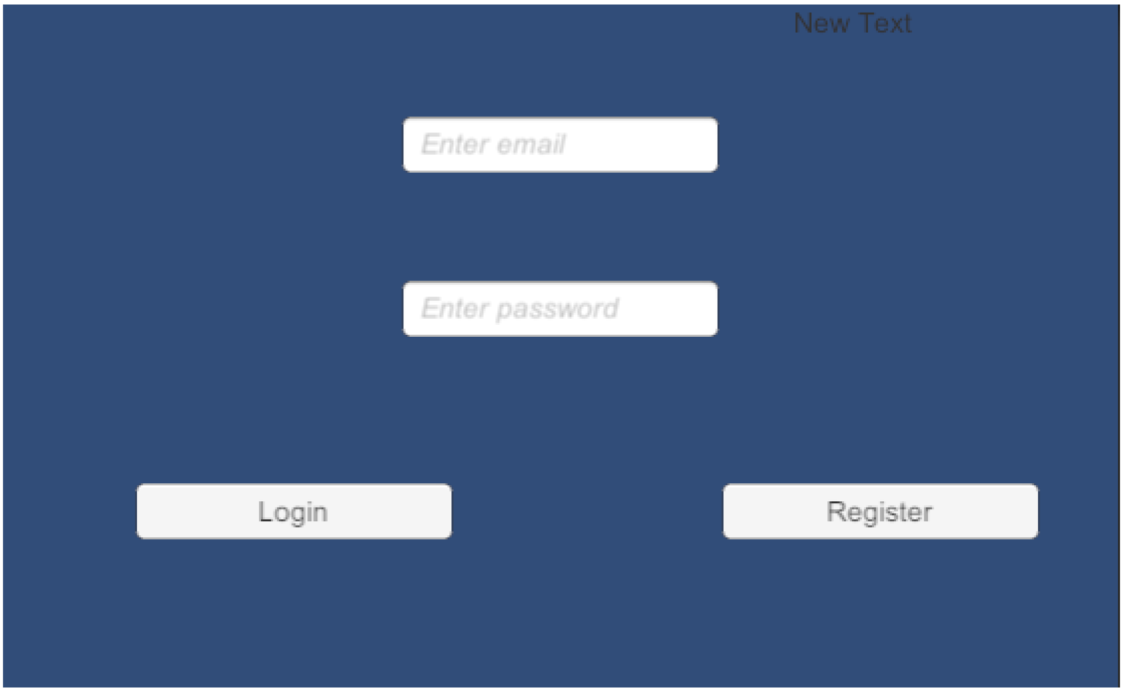


Figure 23 Login & Register Activity

Details: Users need to log in or register using their email and password. If the user chooses to register and the email is valid, the system will send data to firebase and allow the user to successfully register. If the login or registration is successful, the page will allow the user to use the application features. If the player has an error in the email or password, the system will display the error in the Feedback Text, which is the New Text in the upper right corner.

6.2.2 Logout Activity

Figure 24 show the logout activity. The Lobby page of Mind Poker require the user to logout.

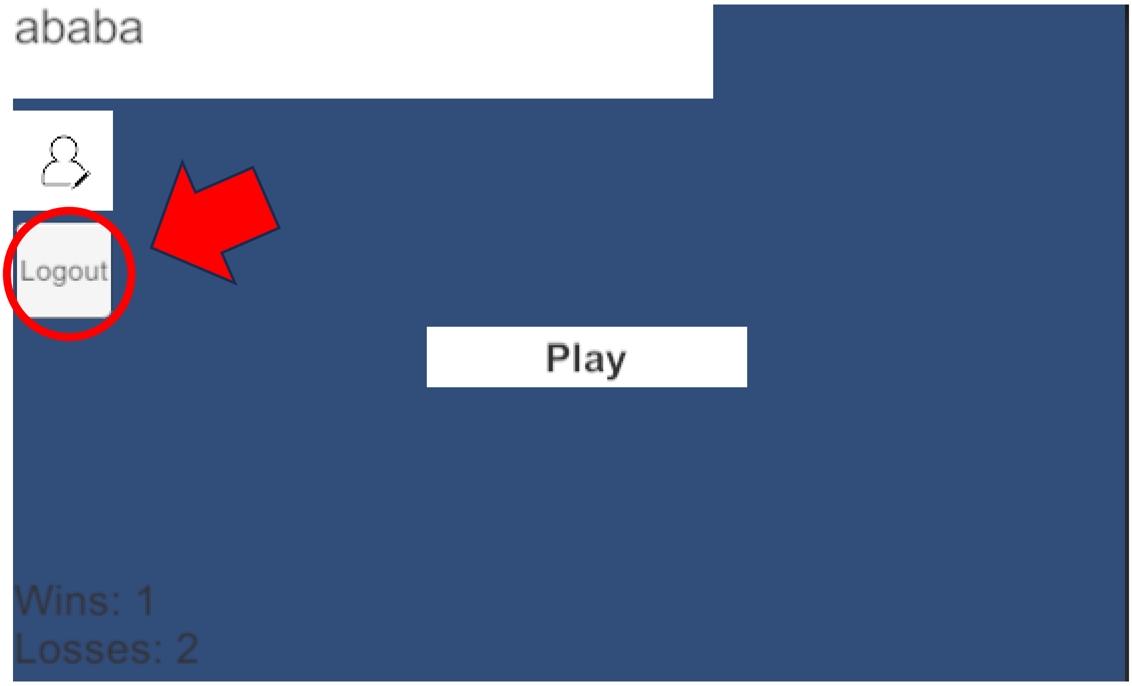


Figure 24 Logout Activity

Details: User can logout account by just click the logout button. After user logout, user will go back to Login Page.

6.2.3 Change username Activity

Figure 25 show the change username activity. The user can change their username in the Lobby page of Mind Poker.

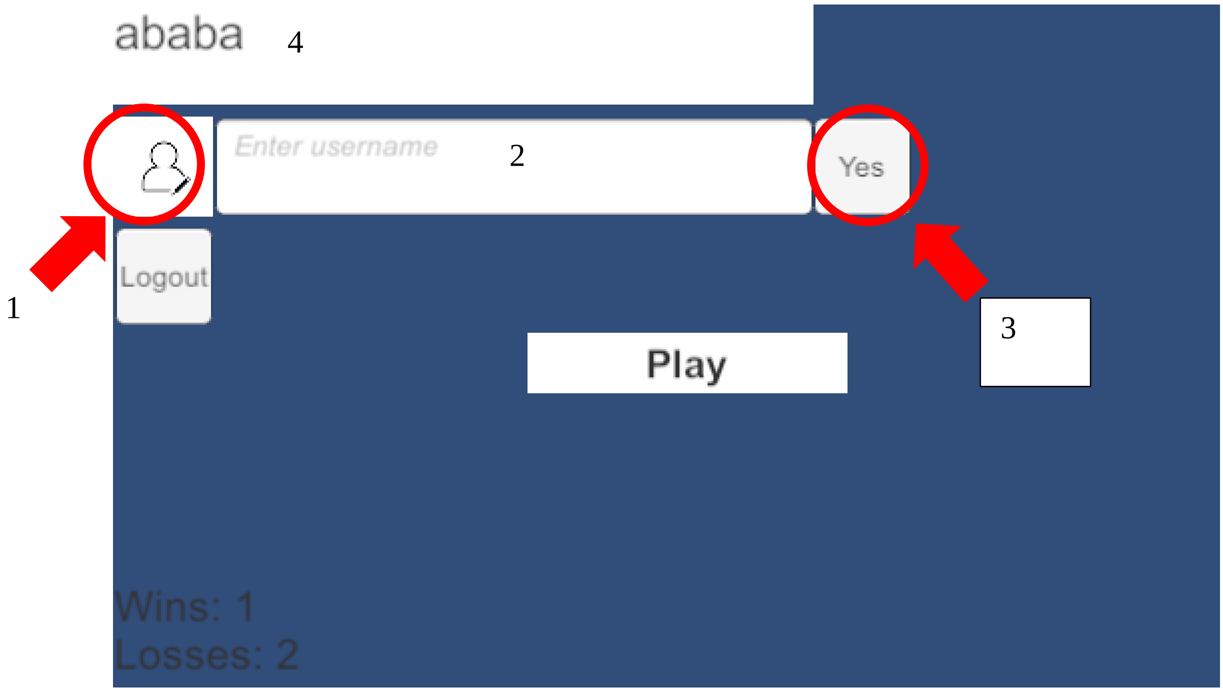


Figure 25 Change username Activity

Details: User can click the change username button, which is the number 1 on the screen in figure 6.2.3. After clicking, the input text (number 2) and the comfirm button (number 3) will appear. The user can enter any name the user wants to change and click the comfirm button to confirm the name change. After the change, the system will change the username in the user database in Firebase and it will appear in the username text (number 4). If the user suddenly does not want to change the username or clicks the change username button by mistake, the player can click the change username button again to close the input text and comfirm button.

6.2.4 View User Win Lose History Activity

Figure 26 show the view user win lose history activity. The user can view their win lose history in the Lobby page of Mind Poker.

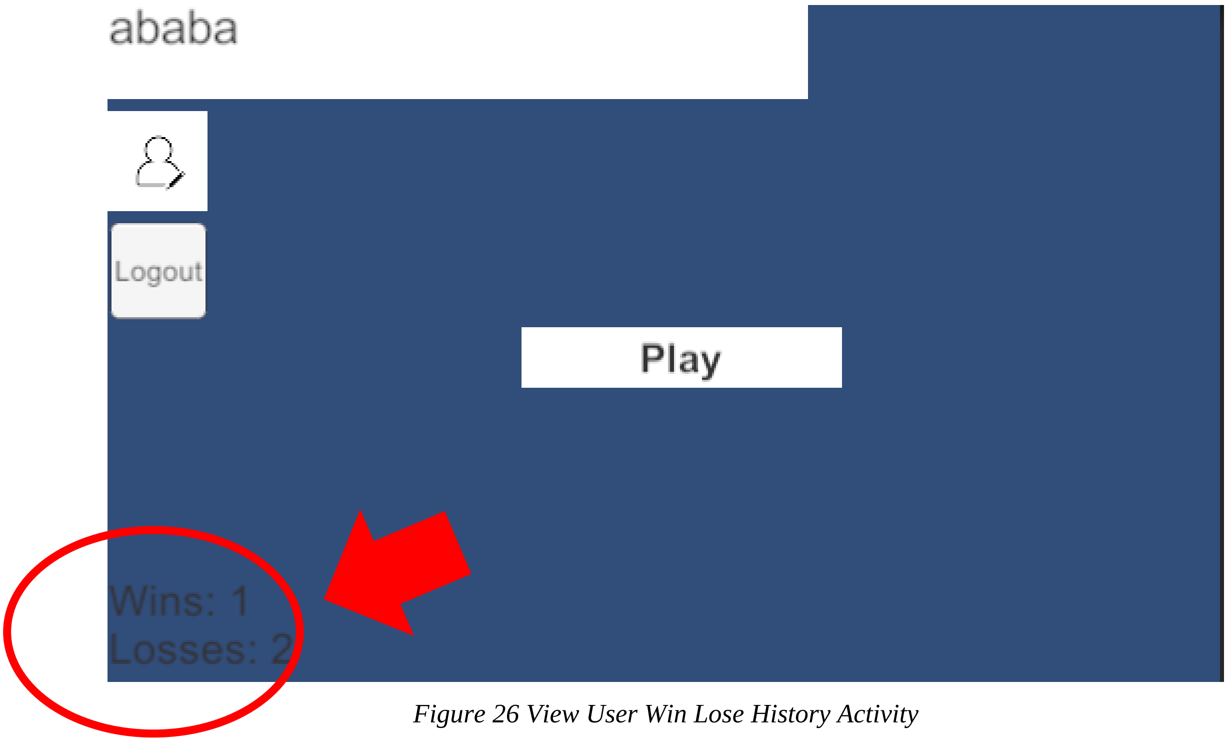


Figure 26 View User Win Lose History Activity

Details: Users can see the detailed win and loss records they have accumulated.

6.2.5 Play Game Activity

Figure 27 show the Play Game activity. The user can start play game in the Lobby page of Mind Poker.

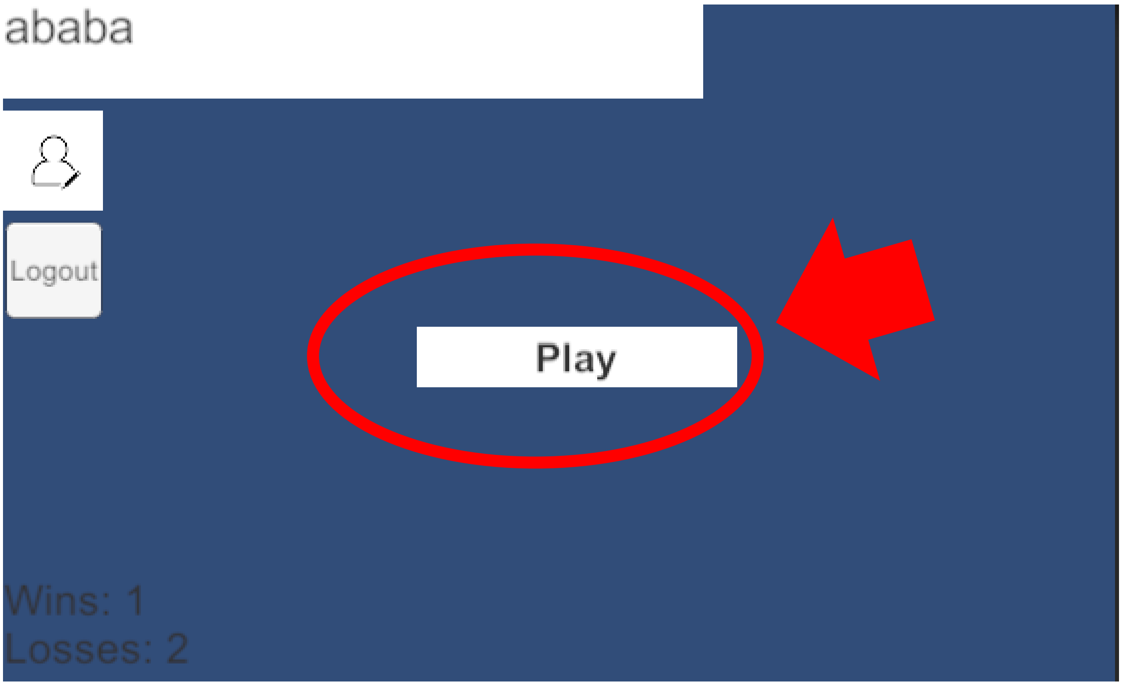


Figure 27 Play Game Activity

Details: User can start play game by click the Play Button. After Click Play Button, user will go to Loading Page for the next step.

6.2.6 Loading & Cancel Activity

Figure 28 show the loading activity. The user need wait to connect in the Loading page of Mind Poker.

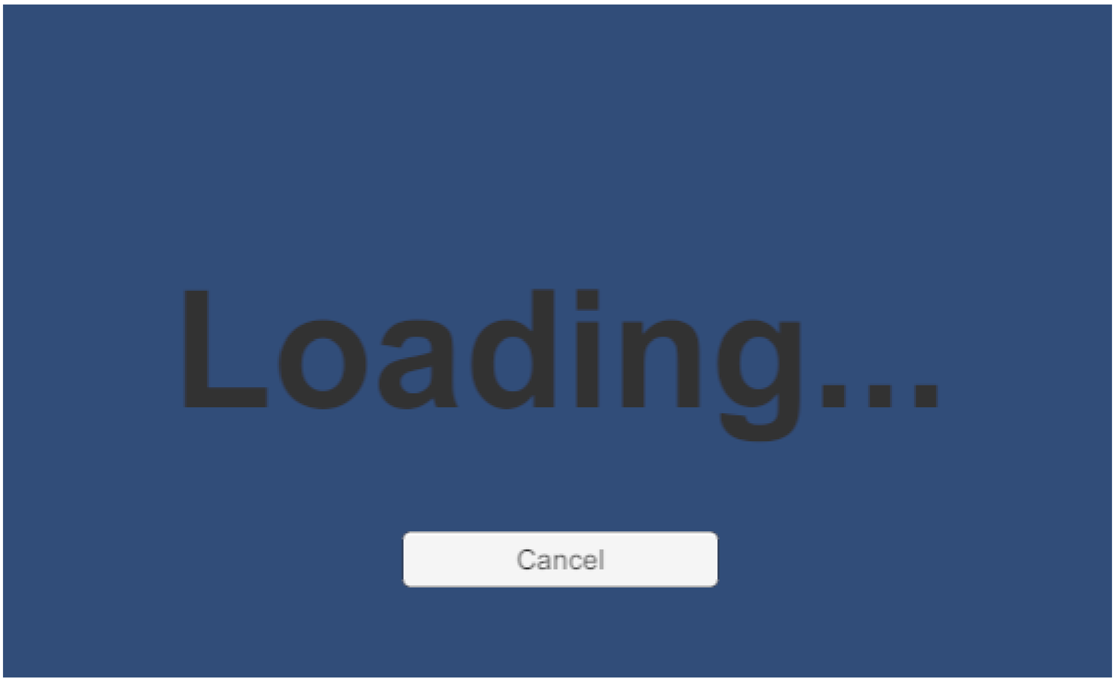


Figure 28 Loading & Cancel Activity

Details: User need to wait for connect so that both users can connect together to play a game. The cancel button is for user to cancel connect the matchmaking.

6.2.7 Game Activity

Below figure will show about what user can do in Game Activity.

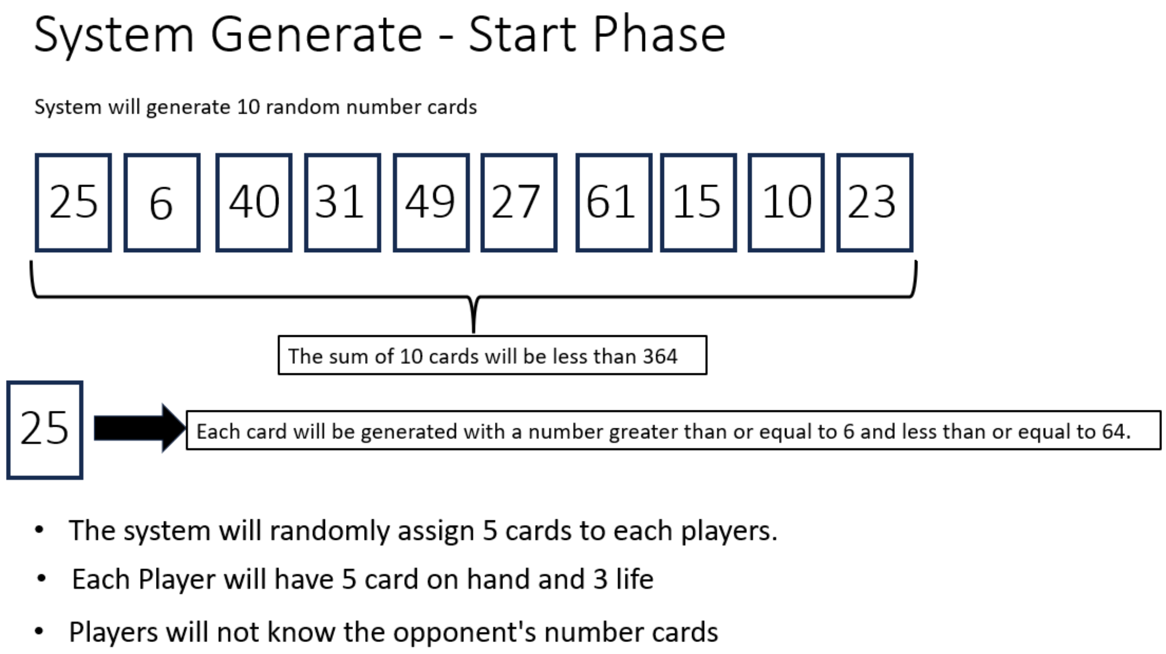


Figure 29 Game Activity-1

Figure 30 Game Activity-2

Pick Card Phase

Player will have 15 seconds to choose number card.

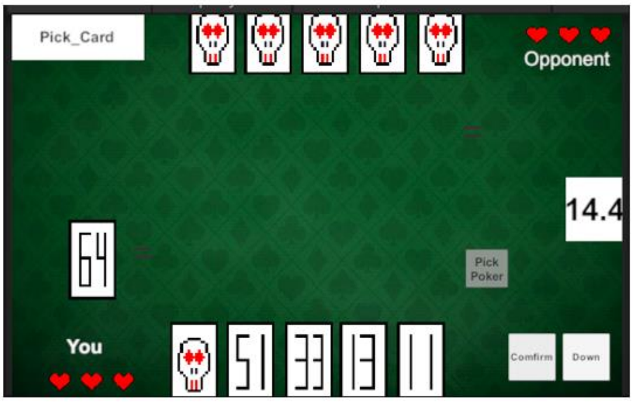


Figure 31 Game Activity-3

Check Number Phase

Player will have 10 seconds to check opponent number card.

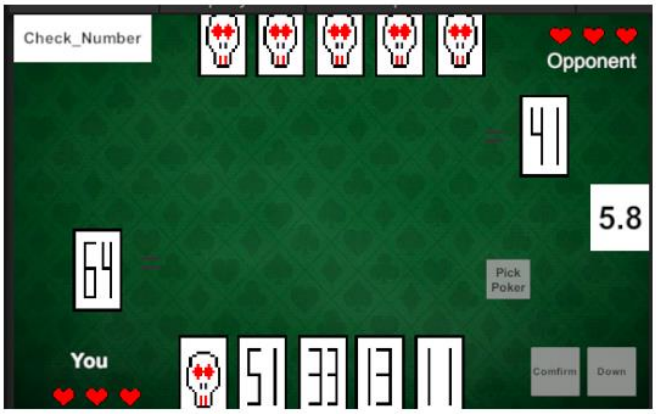


Figure 32 Game Activity-4

Pick Poker Phase

Player need to click Pick Poker button to pick poker

Figure 33 Game Activity-5

Pick Poker Phase

This is what Player will see after click Pick Poker button
After pick 5 poker card, Player need to click Comfirm Button

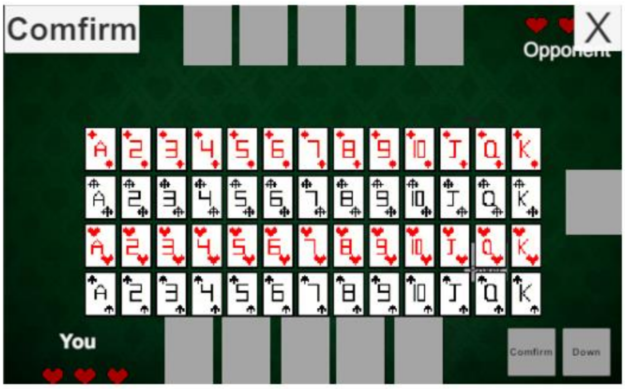


Figure 34 Game Activity-6

Settlement Phase

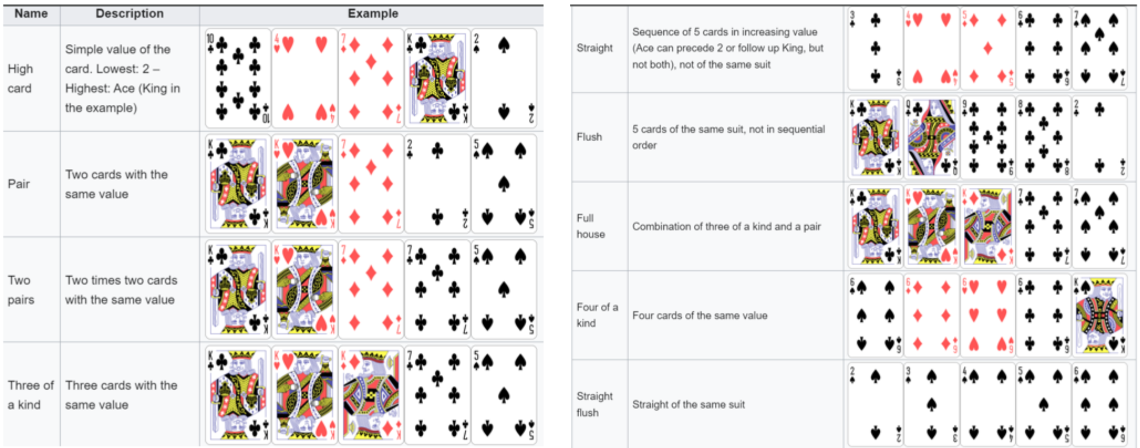
Player will check Oppenent Poker Card

System will calculate the card rank and decide who win in this round.

Figure 35 Game Activity-7

How To Win

The following table shows the possible hand values in increasing order.



Name	Description	Example
High card	Simple value of the card. Lowest: 2 – Highest: Ace (King in the example)	
Pair	Two cards with the same value	
Two pairs	Two times two cards with the same value	
Three of a kind	Three cards with the same value	
Straight	Sequence of 5 cards in increasing value (Ace can precede 2 or follow up King, but not both), not of the same suit	
Flush	5 cards of the same suit, not in sequential order	
Full house	Combination of three of a kind and a pair	
Four of a kind	Four cards of the same value	
Straight flush	Straight of the same suit	

Figure 36 Game Activity-8

How To Win

From Low to High is:

- High Card
- Pair
- Two pairs
- Three of a kind
- Straight
- Flush
- Full house
- Four of a kind
- Straight flush

Figure 37 Game Activity-9

For example:

Because there are no 5 A cards in a deck, we will look at the card pattern at this time.
Player 1 card pattern is “four of a kind”
Player 2 card pattern is “Straight Flush”
So the winner will be P2, because P2 card pattern is higher than P1.

Figure 38 Game Activity-10

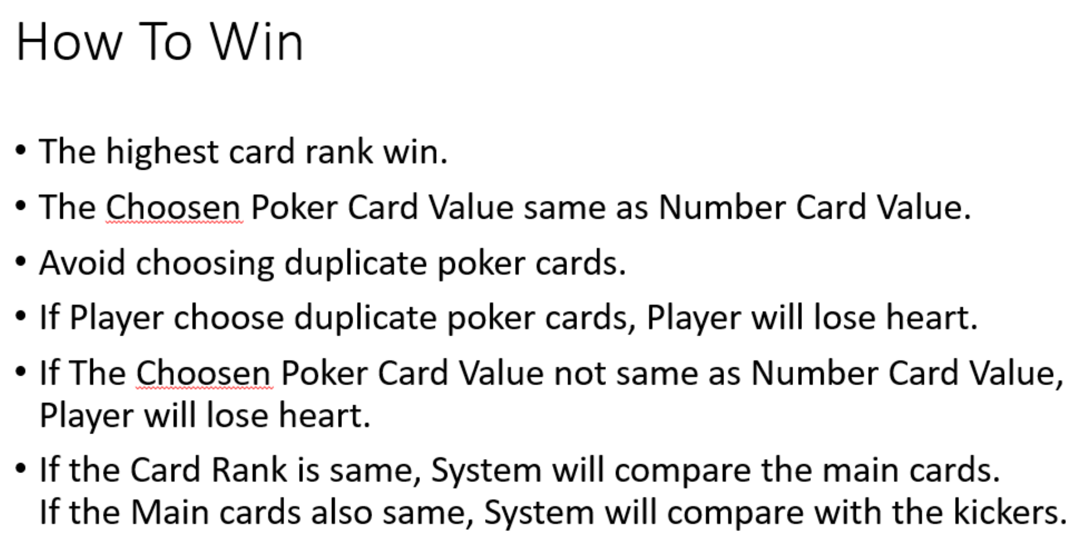


Figure 39 Game Activity-11

Details: Game will end when one of the player hearts is equal to 0. The game phase will repeat until the game ends. The game phase start from Check Card, Pick Card, Check Number, Pick Poker, Settlement and back to Check Card again for a new round.

6.3 Summary

This chapter talk about what user can do in the application and what how the program run. The next chapter is project testing and will discuss the testing of the project.

CHAPTER 7

7 PROJECT TESTING

7.1 Introduction

This chapter will discuss the testing of the project. To deliver a good software product, the developer needs to test the system. This phase is important to find any errors or bugs before releasing it to the user. Software testing ensures the quality of software and represents the final review of other phases of software. By executing this phase, it will help to address known issues and bugs before the product is released.

7.2 Unit Testing

Pre-conditions:

1. The application will be tested by the developer.
2. All tools and software must be tested.

Table 6 Unit Testing

Test Case ID	Test Case	Test Data	Expected Result	Actual Result	Status (Past/Fail)
UT_1	Testing on the platform to run the software	Notebook computer	Condition is good and functions well	Able to carry out all the project functionalities	Pass
UT_2	Testing on computer operating system	Windows operating system	The operating system should be able to run the required software	Required software installed and running properly	Pass
UT_3	Testing on software requirements to run the project	Unity	Should be able to run the application	Unity runs properly	Pass

UT_4	Testing on the network connection	Notebook computer	Notebook computer should accept network signal and show network status	Notebook computer successfully connect to Wi-Fi and receives a network signal	Pass
UT_5	Test out browser functionality	Web Browser (Microsoft Edge)	Able to open Firebase console	Web browser successfully open Firebase console	Pass
UT_6	Determine database functionality	Firebase database	The Firebase database is connected with the project using the project package name, and able to store data	Firebase database successfully connected and data can be stored and retrieved	Pass
UT_7	Determine real-time functionality	PUN 2	PUN 2 is connected with project using the project package name, and able to connect, receive and share data.	PUN 2 successfully connected and data can be received and share.	Pass
UT_8	Testing on software requirements to run the project	Visual Studio Code	Should be able to run the application	Visual Studio Code runs properly	Pass

7.3 Functional Testing

Functional testing is a software testing method that focuses on verifying that the software behaves according to its functional requirements. It involves creating test cases based on the specified requirements to ensure that each feature and function of the software works correctly. During functional testing, inputs are provided to the system, and the outputs are checked to confirm they match the expected results. This type of testing is essential to ensure that the software performs its intended functions accurately and meets the needs of the users.

7.3.1 Login & Register Activity

Table 7 Functional Testing: Login & Register Activity

Test Case ID	**Test Case Name**	**Test Case Description**	**Test Steps**		
			Step	**Expected**	**Test Status (Past/Fail)**
LR_1	Validate Email	To verify that the user can login with using email	Enter the login password and without email, and click the "Login" button	An error message "Invalid email" must be displayed at feedback text.	Pass
LR_2	Validate Email	To verify that the user need to enter valid email	Enter invalid login email and password, and click the "Login" button	An error message "Invalid email" must be displayed at feedback text	Pass
LR_3	Validate Password	To verify that the user needs to enter a password for login	Enter the login email without password and click "Login" button	An error message "Invalid password" must be displayed	Pass

LR_4	Validate Password	To verify that the user needs to enter a password with a minimum of 6 characters	Enter the login email and less than 6 characters password, and click the “Login” button	An error message “Password too short, enter minimum 6 characters” must be displayed	Pass
LR_5	Validate Email	To verify that the user can register with using email	Enter the login password and without email, and click the “Register” button	An error message “Invalid email” must be displayed at feedback text.	Pass
LR_6	Validate Email	To verify that the user need to enter valid email	Enter register email without “@gmail.com” and password and click the “Register” button	An error message “Invalid email” must be displayed at feedback text	Pass
LR_7	Validate Password	To verify that the user needs to enter a password for login	Enter the register email without password and click “Register” button	An error message “Invalid password” must be displayed	Pass
LR_8	Validate Password	To verify that the user needs to enter a password with a minimum of 6 characters	Enter the register email and less than 6 characters password, and	An error message “Password too short, enter minimum 6	Pass

			click the “Register” button	characters” must be displayed	
LR_9	Login Button	To test whether the user goes to the Lobby Page after successfully logging in	Enter the correct login email and password and click the “Login” button	Go to Lobby Page	Pass
LR_10	Register Button	To test whether the user goes to the Lobby Page after successfully register	Enter the correct register email and password and click the “Register” button	Go to Lobby Page	Pass

7.3.2 Logout Activity

Table 8 Functional Testing: Logout Activity

Test Case ID	**Test Case Name**	**Test Case Description**	**Test Steps**		
			Step	**Expected**	**Test Status (Past/Fail)**
LA_1	Logout	To test the logout button is work	After enter the Lobby Page, click the logout button.	Logout Account and go to Login Page	Pass

7.3.3 Change username Activity

Table 9 Functional Testing: Change username Activity

Test Case ID	Test Case Name	Test Case Description	Test Steps		
			Step	**Expected**	**Test Status (Past/Fail)**
CU_1	Open Input Text	To test did the input text and comfirm button will appear after click the change username button	Click “Change username” button	Input Text and Comfirm button appear	Pass

7.3.4 View User Win Lose History Activity

Table 10 Functional Testing: View User Win Lose History Activity

Test Case ID	Test Case Name	Test Case Description	Test Steps		
			Step	**Expected**	**Test Status (Past/Fail)**
WL_1	View Win Lose	In order to see whether the player's win or loss is the same as the firebase database	View the win lose text	Win lose text same with the database data	Pass

7.3.5 Play Game Activity

Table 11 Functional Testing: Play Game Activity

Test Case ID	Test Case Name	Test Case Description	Test Steps		
			Step	**Expected**	**Test Status (Past/Fail)**
PG_1	Play Game	To test do user can go to Loading Page for connect to server	Click the "Play" button	Go to Loading Page	Pass

7.3.6 Loading & Cancel Activity

Table 12 Functional Testing: Loading & Cancel Activity

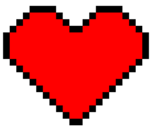
Test Case ID	Test Case Name	Test Case Description	Test Steps		
			Step	**Expected**	**Test Status (Past/Fail)**
LC_1	Connect to Server	To Test do user connect to server so that user can connect with another user	Automatically Connect	Debug log display "connect to photon server"	'Pass
LC_2	Join Room	To let user can connect and play together	Automatically Join	Debug log display "Join Room"	Pass
LC_3	Create Room	If user can't find any room with user inside, user will	Automatically Create	Debug log display "No Room Found, Create Room"	Pass

		create a room and wait another user join			
LC_4	Cancel Matchmaking	Test do user can cancel	Click "Cancel" button	Go back to Lobby Page	Pass

7.3.7 Game Activity

Table 13 Functional Testing: Game Activity

Test Case ID	**Test Case Name**	**Test Case Description**	**Test Steps**		
			Step	**Expected**	**Test Status (Past/ Fail)**
GAME_1	Phase Text	Does Phase Text run according to the setting	Automatically Run	Start From -Check Card -Pick Card -Check Number -Pick Poker -Settlement	Pass
GAME_2	Phase Time	Does Phase Time run according to the setting	Automatically Run	Start From -10s -15s -10s -20s -10s	Pass
GAME_3	Phase Text & Phase Time	Does Phase Text and Phase Time run together	Automatically Run	Phase and Text must match together Check Card-10s Pick Card-15s	Pass

		according to the setting		Check Number-10s Pick Poker-20s Settelment-10s	
GAME_4	Phase Text & Phase Time	Check if both users have the same phase time and the same phase text	Automatically Run	Both user Phase Text & Phase Time will be same	Pass
GAME_5	Heart	Do User heart will change the image from full heart to empty heart after deducting heart	Click "Test_DMG" button	User Heart image will from full heart to empty heart Full Heart:  Empty Heart: 	Pass
GAME_6	Heart	Do User heart will start animation after deducting heart	Click "Test_DMG" button	User Heart will start animation Heart.gif 	Fail
GAME_7	Heart	Do user heart UI will update to another user UI	Click "Test_DMG" button	User 1 heart deduct and user 2 UI heart update	Pass
GAME_8	Draw Number Card	Do the sum of 10 number card value is	Automatically Run	Debug log show "(sum of number card value)	Pass

		lower than 364			
GAME_9	Draw Number Card	Do the number card have display at user number card	Automatically Run	User number card have display the 5 number card	Pass
GAME_10	Draw Number Card	Do the number card have display different at user number card	Automatically Run	User number card is all different	Pass
GAME_11	Draw Number Card	Do both user number card have display at both user number card	Automatically Run	Both user number card have display their own number card	Pass
GAME_12	Draw Number Card	Do both user number card is different	Automatically Run	Both user number card is all different	Pass
GAME_13	Pick Number Card	Do user can pick number card	Choose number card and click “Comfirm” button	The selected number card will go to number card slot	Pass
GAME_14	Pick Number Card	Do number card will have effect when user choose it	Click the number card	Number card will appear line effect	Pass
GAME_15	Pick Down Number Card	After pick the number card to number	Click “Down” button	Number card from the number card	Pass

		card slot, do user can pick down back the number card		slot will back to user number card	
GAME_16	Pick Down Number Card	If number card slot didn’t have number card and user still click “Down” button	Click “Down” button	Debug log “number card slot have no card”	Pass
GAME_17	Number Card Slot	Check do the both user number card slot have number card so that user can go to the next phase	Choose number card and click “Comfirm” button, and wait till the phase end	Both of the user player card slot have number card and change to the next phase	Pass
GAME_18	Number Card Slot	Deduct user heart if user number card slot didn’t have number card	No choose any number card and wait till the phase end	Deduct the user heart that didn’t have number card in the user number card slot	Pass
GAME_19	Number Card Slot	Rerun the phase when one of the user number card slot didn’t have number card	No choose any number card and wait till the phase end	Rerun Pick Card Phase	Pass

GAME_20	Check Number Phase	Do both user number card slot's card has update to another user number card slot UI	Automatically Run	Both user number card slot's card will update to another user number card slot	Pass
GAME_21	Comfirm Button	Do Comfirm Button will only useable at Pick Card Phase	Automatically Run	"Comfirm" button color will turn white and useable at Pick Card Phase	Pass
GAME_22	Down Button	Do Down Button will only useable at Pick Card Phase	Automatically Run	"Down" button color will turn white and useable at Pick Card Phase	Pass
GAME_23	Pick Poker Button	Do Pick Poker Button will only useable at Pick Poker Phase	Automatically Run	"Pick Poker" button color will turn white and useable at Pick Card Phase	Pass
GAME_24	Pick Poker Button	Do Poker Panel will appear	Click "Pick Poker" Button	Appear Poker Pannel	Pass
GAME_25	Poker Panel	Do Poker Panel can close it	Click the Poker Panel's "X" button	Close the Poker Panel	Pass
GAME_26	Poker Panel	Do the poker card can choose	Simply click the poker card	Poker card will become yellow highlighted	Pass

GAME_27	Poker Panel	Do comfirm button work without choosing any card	Click “Comfirm” button	Debug log “need to select poker card”	Pass
GAME_28	Poker Panel	Do comfirm button work if no choose 5 card	Click “Comfirm” button without choosing 5 cards	Debug log “must choose 5 cards”	Pass
GAME_29	Pick Poker	Do Poker Card will appear after choose and click the “comfirm” button	Choose 5 poker card and click “comfirm” button	The choosen poker card display at user poker card slot	Pass
GAME_30	Pick Poker	Do the new poker card will replace the old poker card	Choose 5 poker card and click “comfirm” button After that, choose another 5 poker card and click “comfirm” button again	The poker card slot will renew and display the new poker card	Pass
GAME_31	Pick Poker	Do system will calculate the poker sum value, hand	Choose 5 poker card and click “comfirm” button	Debug log “Player choosen poker cards: ~” “Total Value:~” “Hand rank： ~“	Pass

		rank, and the kickers		"Main Card: ~" "Kickers: ~"	
GAME_32	Poker Card slots	Check do the both user poker card slots have poker card so that user can go to the next phase	Choose poker card and click "Comfirm" button, and wait till the phase end	Both of the user player poker slot have poker card and change to the next phase	Pass
GAME_33	Poker Card slots	Deduct user heart if user poker card slot didn't have poker card	No choose any poker card and wait till the phase end	Deduct the user heart that didn't have poker card in the user poker card slot	Pass
GAME_34	Poker Card slots	Rerun the phase when one of the user poker card slot didn't have poker card	No choose any poker card and wait till the phase end	Rerun Pick Poker Phase	Pass
GAME_35	Settlement Phase	Do both user poker card slot's card has update to another user poker card slot UI	Automatically Run	Both user poker card slot's card will update to another user poker card slot	Pass
GAME_36	Settlement Phase	Check do user poker card total value is	Automatically Run	Deduct user heart if the poker card total value is not same as	Pass

		same as user number card value		user number card value	
GAME_37	Settlement Phase	Check Used Poker card	Automatically Run	Deduct user heart if user have used the poker card that previous used before	Pass
GAME_38	Settlement Phase	Compare Poker hand rank	Automatically Run	Compare both user poker hand rank and deduct the user heart with the lower rank	Pass
GAME_39	Settlement Phase	Stop other check if user poker card total value is not same as user number card value	Automatically Run	One of the user poker card total value is not same as user number card value, stop check used card and poker hand rank.	Pass
GAME_40	Settlement Phase	Stop other check if user didn’t pass Check Used Poker card	Automatically Run	One of the user have used the poker card that previous used before, stop compare poker hand rank	Pass
GAME_41	Settlement Phase	Does both user poker card have been recorded	Automatically Run	The poker card have been record	Pass
GAME_42	Settlement Phase	Does both user number card slot have	Automatically Run	Both user number card slot is been clear and no any	Pass

		been clear and both user poker card slot have been clear		number card inside. Both user poker card slots is been clear and no any poker card inside	
GAME_43	Heart	When one of the user’s heart equal to zero, do the result will pop up	Automatically Run	One of the user’s heart is equal to 0 and the result pop up. The user that no hearts result is “You Lose!!!” and the user that still have hearts result is “You Win!!!”	Pass
GAME_44	Result	Player can go back to lobby page when the game is end	Click “Back to Lobby” button	After the result, player can go back to Lobby Page	Pass

7.4 System Testing

Pre-conditions:

1. The application will be tested by the developer.
2. The integrated modules will be tested as a whole system.

Table 14 System Testing

Test Case ID	Test Case	Test Case Data	Expected Result	Actual Result	Test Status (Past/Fail)
SS_1	Check the application as a whole system	Application, Wi-Fi signal	Run the system. Sign In Activity will be shown	Application show Sign In Activity as mentioned	Pass
SS_2	Allow user to enter data	Input field, upload image button and spinner	The user can enter data, upload image and click on the spinner to choose an option	The user can enter data for all input fields	Pass
SS_3	Verify data in database same as user input	Firebase database	Data should be sent to database according to attribute name	Data exist in database with correct attributes name	Pass
SS_4	Do user can send and receive game data	PUN 2	Game data should send and receive each other so can achieve	The user can receive and send game data each other	Pass

	in real-time		the real-time environment		

7.5 Summary

Chapter 7 emphasizes the significance of testing in the software development process. Testing is crucial for identifying and rectifying errors or bugs before the software is released to users. This chapter particularly focuses on functional testing, a method that verifies the software's functionality against its specified requirements. By simulating real-world inputs and comparing the outputs to expected results, functional testing ensures that the software performs as intended and meets user needs. This phase is essential to guarantee the quality and reliability of the software before its deployment.

Chapter 8, which deals with the summarization of the project, builds on the insights gained from the testing phase. The findings from functional testing, along with other phases, will inform the overall assessment of the project. In summarizing the project, Chapter 8 will reflect on how well the software meets its objectives, the effectiveness of the testing strategies employed, and the readiness of the software for release. The insights from Chapter 7 will be crucial in evaluating the success of the project and ensuring that it is delivered as a high-quality product..

CHAPTER 8

8 CONCLUSION

8.1 Project Summarization

This project aims to develop a card game system called "Mind Poker" that allows two players to engage simultaneously in a virtual poker game. The system is designed to provide a seamless and engaging experience for players, simulating the real-world dynamics of a poker game while incorporating the flexibility and convenience of a digital platform.

The development of this project involved carefully designing the game mechanics and ensuring that all interactions between players are accurately reflected on their respective user interfaces. To achieve this, the system's architecture includes mechanisms for managing the cards, handling player inputs, and ensuring that game rules are enforced consistently.

Functional testing played a critical role in the development process, as it was used to validate that each feature of the game worked as intended. This testing phase helped identify and address any issues or bugs, ensuring that the final product meets the expectations of the users and provides a reliable gaming experience.

Each aspect of the project, from the initial design to the final testing, was crucial in ensuring the overall success of "Mind Poker." The attention to detail in each phase—design, development, and testing—has contributed to delivering an interactive and enjoyable poker experience for two players. The project's success highlights the importance of a comprehensive approach, where every component plays a vital role in creating a high-quality software product.

In conclusion, "Mind Poker" successfully delivers an interactive and enjoyable poker experience for two players. The system's design and testing processes have ensured that the game is both functional and user-friendly, making it ready for release. This project demonstrates the importance of thorough testing and careful design in delivering a high-quality software product.

8.2 Future Enhancement

There are several enhancements planned for "Mind Poker" to further improve the game's experience and functionality.

- Ranking System: A ranking system will be implemented to allow players to see their standings relative to others. This feature will add a competitive edge to the game, encouraging players to improve their skills and climb the leaderboard.

- UI Redesign: The overall appearance of the application will be redesigned to make it more visually appealing. This enhancement will focus on modernizing the interface, improving user experience, and ensuring that the game is both aesthetically pleasing and easy to navigate.

- Optimization: Performance optimization is another key area of focus. By refining the underlying code and system architecture, the game will run more smoothly, providing a better experience, especially on lower-end devices.

- Bug Fixes: Continuous improvement will be ensured by addressing any bugs that are identified post-release. Regular updates will be provided to resolve issues, ensuring a stable and reliable gaming experience for all users.

- Monetization Features: I will also add a krypton gold system and many projects that can make money. A monetization system will be introduced, offering various in-game purchases and features that can generate revenue. This may include cosmetic items, special cards, or premium features that enhance gameplay while offering players additional customization options.

In summary, these future enhancements will not only improve the gameplay experience for current users but will also attract new players, making "Mind Poker" a more engaging and successful product.

REFERENCES

Usogui (嘘喰い, lit. 'The Lie Eater') is a Japanese manga series written and illustrated by Toshio Sako [ja]. It was serialized in Shueisha's *seinen* manga magazine *Weekly Young Jump* from May 2006 to December 2017, with its chapters compiled into 49 *tankōbon* volumes. The manga was later adapted into original video animation, which was released on October 19, 2012.[2]

Usogui - Wikipedia

Yu-Gi-Oh! Master Duel is a free-to-play digital collectible card game based on the *Yu-Gi-Oh! Trading Card Game*, developed and published by Konami for Microsoft Windows, Nintendo Switch, PlayStation 4, PlayStation 5, Xbox One, Xbox Series X/S, Android, and iOS.[1]

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Texas hold 'em (also known as **Texas holdem**, **hold 'em**, and **holdem**) is one of the most popular variants of the card game of poker. Two cards, known as hole cards, are dealt face down to each player, and then five community cards are dealt face up in three stages. The stages consist of a series of three cards ("the flop"), later an additional single card ("the turn" or "fourth street"), and a final card ("the river" or "fifth street"). Each player seeks the best five-card poker hand from any combination of the seven cards: the five community cards and their two hole cards. Players have betting options to check, call, raise, or fold. Rounds of betting take place before the flop is dealt and after each subsequent deal. The player who has the best hand and has not folded by the end of all betting rounds wins all of the money bet for the hand, known as the pot. In certain situations, a "split pot" or "tie" can occur when two players have hands of

equivalent value. This is also called "chop the pot". *Texas hold 'em* is also the *H* game featured in HORSE and HOSE.
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https://unity.com/cn

APPENDIX A
PROGRESS REPORT

Student Name	:	CALVIN NG ZHI QI
Student ID No	:	2290228-DCS
Programme	:	DIPLOMA IN COMPUTER SCIENCE (DCS)
Supervisor's Name	:	MS SUHAILAH BINTI FAUZI
Project Title	:	ONLINE TWO PLAYER POKER GAME – MIND POKER

Week	Date	Descriptions	Remarks
1	13/5/2024 – 17/5/2024	Reading/thinking of the project	
2	20/5/2024 – 24/5/2024	Research on the Internet was conducted to gain ideas on the project development.	
3	27/5/2024 – 31/5/2024	Research on the Internet was conducted to gain ideas on the project development. Rethinking how to implement the reference game	
4	3/6/2024 – 7/6/2024	Supervisor nomination was conducted. Meeting with main supervisor, for confirming the project title, problem statement, objectives, and methodology used, the feasibility and innovation of the project can be assured. In addition, writing for the project proposal report.	Change the project title
5	10/6/2024 – 14/6/2024	Started to try to create some functions.	

		Because it is the first time to make a game, I need more time to watch videos and tutorials	
6	17/6/2024 – 21/6/2024	Start to design the game UI and change some details and rules of the game	
7	24/6/2024 – 28/6/2024	Start handcrafting the design of each card and testing whether the card can be put into the game	
8	1/7/2024 – 5/7/2024	Start testing while making the game Make details and changes about UI	
9	8/7/2024 – 12/7/2024	Start making card algorithms, poker algorithms, and test errors Go see my supervisor.	Supervisor say keep do
10	15/7/2024 – 19/7/2024	Start connecting to the Firebase database system to check if there is any problem with the database system. Because I am not familiar with it, I have just started looking for methods and tools to achieve online connection.	
11	22/7/2024 – 26/7/2024	Test various links about PUN 2 Start to make how to send and receive data between two online users	
12	29/7/2024 – 2/8/2024	Slowly getting familiar with how to connect 2 users Testing while making	
13	5/8/2024 – 9/8/2024	Finish the application project and give supervisor to preview	Teacher say OK. Need Flow Chart to understand all the function
14	12/8/2024 – 16/8/2024	Writing Final Report and poster.	
15	19/8/2024 – 23/8/2024	Completion of project report. Submission of final project.	

Prepared by,

Signature

(CALVIN NG ZHI QI)

Verified by,

Signature

(MS SUHAILAH BINTI FAUZI)